
Monocular Depth Estimation for Agricultural Machinery: LiDAR-Based Ground Truth Generation and Model Benchmarking

Master's Thesis

Collaboration with AGCO corporation and Aalborg University

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Computer Engineering - AI, Vision and Sound



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Abstract:

Depth perception is essential for safe and efficient operation of agricultural machinery, where vehicles often work in close proximity to terrain boundaries, obstacles, and human workers. Although LiDAR sensors provide accurate depth measurements, their cost and system complexity motivate the use of monocular depth estimation as a camera-based alternative. This thesis studies the performance of modern monocular depth estimation models on real agricultural data using sparse LiDAR-derived depth as supervision. A dataset preparation pipeline is developed to transform raw ROS bag recordings into rectified RGB images paired with projected LiDAR depth maps, including sensor alignment and quality control. Based on this dataset, MonoDepth2, ZoeDepth, and Depth Anything v3 are benchmarked under a unified training and evaluation setup with fixed bag-level splits. Experiments show that sparse LiDAR-supervised fine-tuning substantially improves metric depth accuracy and domain adaptation compared to zero-shot inference, while also revealing that evaluation protocol and ground-truth representation strongly influence reported metrics. The results demonstrate that carefully prepared agricultural datasets enable reliable benchmarking of monocular depth models and provide a solid geometric basis for future safety-oriented applications such as distance estimation and time-to-collision analysis.

Contents

1	Introduction	1
1.1	Problem Statement	3
1.2	Research Questions	3
1.3	Research Hypothesis	3
1.4	Learning Goals	4
1.5	Thesis Structure	4
2	Literature Review	6
2.1	Monodepth2: Self-Supervised Monocular Depth Estimation	6
2.2	ZoeDepth: Metric Monocular Depth Estimation	7
2.3	Depth Anything 3: Unified Geometry and Monocular Depth Estimation . .	8
3	Dataset Preparation	10
3.1	Data Preparation Pipeline	10
3.2	Dataset Engineering	10
3.2.1	Data Preparation and Dataset Generation	11
3.2.2	Depth Map Generation	11
3.2.3	Depth Smoothing	12
3.2.4	Dataset Splitting	13
3.3	LiDAR–Camera Projection	13
3.3.1	LiDAR Point Representation	13
3.3.2	Transformation from LiDAR Frame to Camera Frame	13
3.3.3	Projection onto the Image Plane	14
3.3.4	Depth Assignment and Z-Buffering	14
3.3.4.1	Z-buffering explanation	14
3.3.5	Depth Map Representation	15
3.3.6	Manual Extrinsic Refinement	15
3.4	Dataset Construction and Variants	15
3.4.1	Dataset Structure and Sample Definition	16
3.4.2	Bag-Level Dataset Splitting	16
3.4.3	Sparse Ground-Truth Dataset	16
3.4.4	Smooth Ground-Truth Dataset	17
3.4.5	Depth Value Representation and Scaling	17
3.4.6	Dataset Usage in Training and Evaluation	17

4	Model Training and Evaluation	19
4.1	Model Training Strategy	19
4.1.1	Training Overview	19
4.1.2	Depth Prediction	20
4.1.3	Valid Pixel Masking	20
4.2	Loss Functions	20
4.2.1	Masked Absolute Depth Loss	21
4.2.2	Masked Logarithmic Depth Loss	21
4.2.3	Edge-Aware Smoothness Regularization	21
4.2.4	Total Training Loss	22
4.2.5	Training Procedure	22
4.2.6	Models Used in Experiments	23
4.3	Evaluation Metrics	23
4.3.1	Evaluation Setup and Notation	23
4.3.2	Absolute Relative Error (Abs Rel)	24
4.3.3	Squared Relative Error (Sq Rel)	24
4.3.4	Root Mean Squared Error (RMSE)	25
4.3.5	Logarithmic RMSE (RMSE Log)	25
4.3.6	Threshold Accuracy Metrics ($\delta_1, \delta_2, \delta_3$)	25
5	Experiments	27
5.1	Experimental Setup	27
5.1.1	Dataset Usage and Splits	27
5.1.2	Ground-Truth Depth Variants	28
5.1.3	Training Configuration	28
5.1.4	Hardware and Software Environment	28
5.1.5	Evaluation Protocol	29
5.2	Datasets Used in Experiments	29
5.3	Training Experiments	30
5.3.1	Experiment Groups	30
5.3.2	Baseline (Zero-shot) Runs	30
5.3.3	Monodepth2: Convergence and Validation Trends	31
	Sparse supervision (Monodepth2).	31
	Smooth supervision (Monodepth2).	31
	Training and validation curves.	33
5.3.4	ZoeDepth: Convergence and Validation Trends	34
	Sparse supervision (ZoeDepth).	34
	Smooth supervision (ZoeDepth).	34

Training and validation curves.	36
5.3.5 Depth Anything v3: Convergence and Validation Trends	37
Sparse supervision (DA-v3).	37
Smooth supervision (DA-v3).	37
Training and validation curves.	39
5.3.6 Training Behaviour	40
5.4 Quantitative Results	40
5.4.1 Baseline vs. Sparse LiDAR-Supervised Fine-tuning	40
Zero-shot baseline.	40
Impact of sparse LiDAR fine-tuning.	41
Effect of dataset fraction.	41
5.4.2 Fine-tuning on Smooth Depth and Evaluation on Smooth Depth . .	41
Monodepth2 remains stable.	41
ZoeDepth and DA-v3 show inflated relative-error metrics. . .	42
5.4.3 Diagnostic Evaluation: Smooth-trained Models Evaluated on LiDAR- GT Pixels	42
5.4.4 Diagnostic Evaluation: Cropped-Region (“1.4/3 of the image”) . . .	43
5.5 Model Comparison and Discussion	43
5.5.1 Qualitative Comparison: Zero-shot vs Fine-tuned Predictions	43
5.5.2 Model-Specific Behaviour	45
5.5.2.1 Monodepth2	45
5.5.2.2 ZoeDepth	48
5.5.2.3 Depth Anything v3	48
5.5.3 Cross-Model Trends	50
5.5.4 Error Analysis and Experimental Design Choices	50
5.5.4.1 Why Use a Cropped Evaluation Region (1.4/3 of the Image)	50
5.5.4.2 Why Perform Multiple Evaluation Configurations	50
5.5.4.3 Implications for Practical Use	51
6 Conclusion and Future Work	52
6.1 Main Contributions	52
6.2 Key Findings and Limitations	53
6.3 Future Work	53
6.3.1 Downstream Use Case: Depth-Based Proximity Cues (Concept) . .	53
6.3.2 Downstream Use Case: TTC Estimation from Depth (Concept) . . .	54
Bibliography	55

1 Introduction

Depth perception is really important for agricultural machinery to work safely and smartly. These machines are used outside in environments where it is crucial to know exactly how far away people, objects, crops, and the ground are. This is necessary to avoid accidents and tasks and to make good decisions. When we have information about depth, we can do things like detect obstacles, keep an eye on human safety, analyze the terrain, and navigate through fields. Depth perception is essential for all these tasks in fields, and it helps with depth information that we need to work safely and efficiently.

Traditionally people get depth information using sensors like LiDAR. These sensors give us accurate measurements of how far away things are in 3D space. Lidar sensors are really expensive, and they make a lot of data that can be hard to handle. They also make the whole system more complicated. On the other hand, regular cameras that just capture color pictures are cheap and easy to use. They are also very light and simple to set up. The problem with these cameras is that they do not directly tell us how far away things are. LiDAR sensors are good at measuring depth, but regular cameras are not. This problem has led to a lot of research on figuring out how deep things are in a picture using one regular color photo. The goal of this research is to create a map of how far away things are from the camera, and this is really useful for the machines that work in the fields and the agricultural machinery that goes over the fields.

Deep learning has really improved how well we can figure out how deep things are with one camera. Some models like Monodepth2 [1], ZoeDepth [2], and Depth Anything v3 [3] have worked well with lots of different datasets and environments. These datasets do not really show what things look like or the problems we face in farming areas. There are things that make it harder to understand farming areas, such as light that changes, plants, dust, ground that is not even, and big open spaces. These things add a lot of complexity to depth estimation.

Another big challenge is the availability of reliable ground-truth depth data. While LiDAR provides accurate depth measurements, raw LiDAR–camera recordings cannot be used directly for training or evaluation. The data must be carefully synchronized, projected, filtered, and aligned to generate consistent depth maps. In practice, this data preparation step is often application-specific and lacks a standardized workflow, making systematic evaluation across models difficult.

This thesis focuses on combining LiDAR-based supervision with monocular depth estimation for agricultural machinery using real multi-sensor data collected in collaboration with AGCO. A key contribution is the development of an automated and reproducible pipeline that converts raw ROS bag recordings into training-ready datasets, including Li-

DAR–camera projection, depth map generation, and quality control. Based on this prepared dataset, multiple state-of-the-art monocular depth estimation models are benchmarked under consistent training and evaluation conditions. In addition, controlled experiments with varying amounts of training data provide insight into how dataset size influences overall behaviour and model stability in an industrial agricultural context.

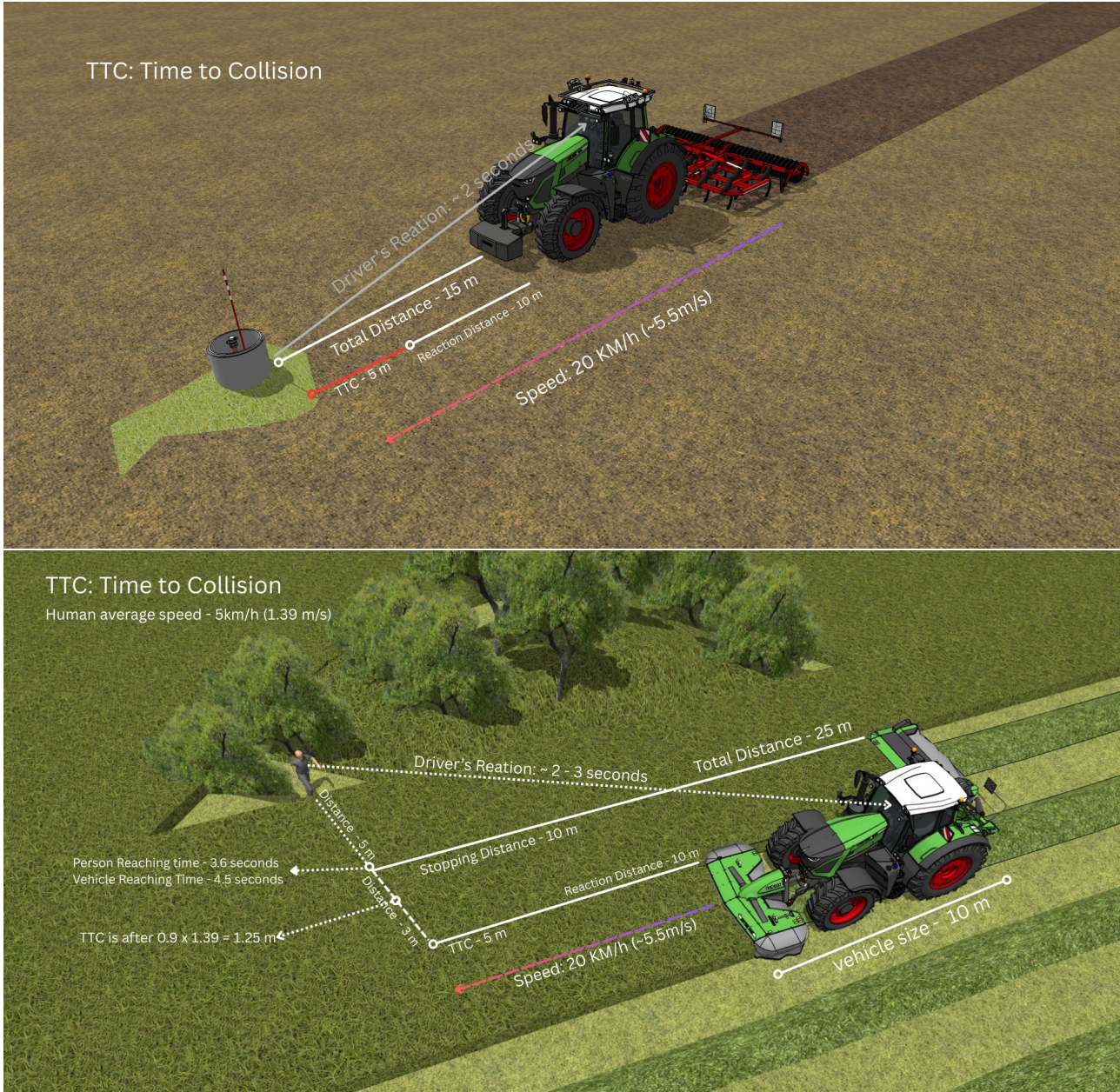


Figure 1.1: Example agricultural scenarios motivating monocular depth estimation for obstacle proximity and time-to-collision (TTC) reasoning.

Figure 1.1 highlights some safety-critical scenarios in agricultural settings where depth perception is critical for judging the nearness of objects and time-to-collision (TTC) scenarios. In these examples, tractors operate in close proximity to static obstacles, vegetation boundaries, and human workers, often with limited reaction time and changing terrain conditions. Reliable depth estimation enables the system to reason about how far an obstacle is

and how quickly the vehicle is approaching it, which is a fundamental requirement for collision avoidance, warning generation, and emergency braking. These scenarios motivate the use of monocular depth estimation as a key perception component for future safety-oriented agricultural machinery, even when expensive active sensors such as LiDAR are unavailable or limited.

1.1 Problem Statement

Monocular depth estimation models require well-prepared and reliable depth supervision, yet raw LiDAR–camera data cannot be directly used due to sparsity, misalignment, and noise. There is no standardized and automated approach for preparing depth estimation datasets from ROS bag recordings and evaluating modern monocular depth estimation models on industrial agricultural data. As a result, it remains unclear how effective current models are and how LiDAR-based supervision can best improve monocular depth estimation performance in agricultural scenario.

1.2 Research Questions

- How does the dataset need to be prepared?
- Are the selected models sufficient for benchmarking monocular depth estimation performance?
- How can LiDAR data help train better monocular depth estimation models?
- Can an automated pipeline reduce manual work while still producing reliable results?
- Which depth estimation models perform best on company and public datasets?

1.3 Research Hypothesis

This thesis is based on the hypothesis that reliable monocular depth estimation for agricultural machinery can be achieved through effective dataset preparation and LiDAR-supervised fine-tuning, even when only sparse LiDAR measurements are available. It is hypothesized that an automated and reproducible pipeline for processing raw LiDAR–camera recordings into training-ready datasets enables fair and meaningful benchmarking of modern monocular depth estimation models in agricultural environments. Furthermore,

incorporating LiDAR-based supervision is expected to significantly improve metric accuracy, stability, and domain adaptation compared to zero-shot performance, while systematic evaluation under controlled conditions reveals consistent performance trends and practical trade-offs between model architectures. Finally, it is assumed that monocular depth maps trained with LiDAR supervision can serve as a reliable geometric foundation for future safety-oriented applications, such as obstacle proximity estimation and time-to-collision reasoning, even though such downstream systems are not implemented or validated within the scope of this thesis.

1.4 Learning Goals

The learning goals of this thesis are threefold.

- **First**, the thesis aims to develop a strong practical understanding of dataset engineering for monocular depth estimation using real multi-sensor data. This includes learning how to process raw ROS bag recordings, perform LiDAR–camera projection, generate consistent depth supervision, and design automated pipelines that reduce manual intervention while maintaining data quality.
- **Second**, the thesis seeks to analyze and understand the behaviour of state-of-the-art monocular depth estimation models under LiDAR-supervised fine-tuning. This involves studying convergence behaviour, sensitivity to dataset size, and robustness to different ground-truth configurations, as well as identifying strengths and limitations of each model in agricultural environments.
- **Third**, the thesis aims to establish a rigorous and transparent evaluation framework for monocular depth estimation on industrial agricultural data. By applying consistent dataset splits, supervision strategies, and evaluation protocols, the work provides insight into how evaluation design influences reported performance and how quantitative metrics should be interpreted in safety-relevant scenarios.

Together, these learning goals support the broader objective of enabling reliable depth perception for future agricultural machinery, where accurate geometric understanding is a prerequisite for higher-level safety and autonomy functions.

1.5 Thesis Structure

The remainder of this thesis is organized as follows also showed in Figure 1.2. Chapter 2 reviews related work in monocular depth estimation and introduces the state-of-the-art

models considered in this study. Chapter 3 presents the dataset preparation methodology, including the data processing pipeline, LiDAR–camera projection, depth map generation, and dataset construction. Chapter 4 describes the model training and evaluation methodology, covering the training strategy, loss functions, model configurations, and evaluation metrics. Chapter 5 reports the experimental setup and benchmarking results across different models, supervision strategies, and dataset sizes, followed by a detailed comparative discussion. Finally, Chapter 6 summarizes the main conclusions of the thesis and outlines key limitations and directions for future work.

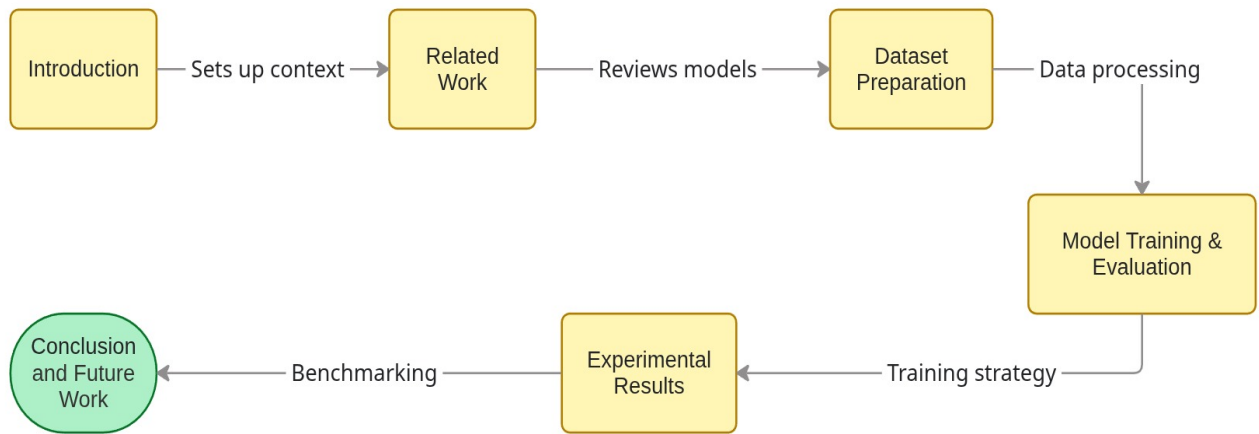


Figure 1.2: Block diagram of thesis structure

2 Literature Review

Monocular depth estimation has been an active research area for many years, driven by the goal of recovering three-dimensional scene structure from a single RGB image. Early approaches relied on handcrafted features and geometric assumptions, while modern methods are dominated by deep learning. Recent advances include self-supervised learning from video sequences, supervised learning with metric depth, and large-scale pretraining using diverse real and synthetic datasets. These approaches differ in supervision requirements, architectural complexity, and generalization ability.

In this thesis, the literature review focuses on three representative monocular depth estimation models: Monodepth2 [1], ZoeDepth [2], and Depth Anything v3 [3]. These methods are selected because they reflect distinct and widely used design paradigms in the field. Monodepth2 represents self-supervised learning without explicit depth labels [1], ZoeDepth targets metric depth prediction using supervised learning [2], and Depth Anything v3 leverages large-scale pretraining and transformer-based architectures [3]. Together, they provide a meaningful and balanced basis for benchmarking monocular depth estimation performance in agricultural environments.

2.1 Monodepth2: Self-Supervised Monocular Depth Estimation

Monodepth2 is a well-established monocular depth estimation method that predicts depth from a single RGB image without relying on ground-truth depth labels. Instead, it uses self-supervision obtained from image sequences or stereo image pairs, making it suitable for large-scale and real-world datasets where collecting depth annotations is difficult. During training, the model learns depth by utilizing geometric consistency between consecutive frames and by reconstructing one image from another using predicted depth and camera motion. This formulation allows the model to infer depth from appearance changes over time. [1]

A key strength of Monodepth2 is that it avoids the need for expensive depth sensors during training. This makes it a strong baseline for outdoor environments, including agricultural scenarios where dense depth annotations are often unavailable. However, since the method primarily learns relative depth, additional scaling is required to recover metric distances at inference time. Despite this limitation, Monodepth2 remains a widely used reference model due to its simplicity, robustness, and strong generalization properties. [1]

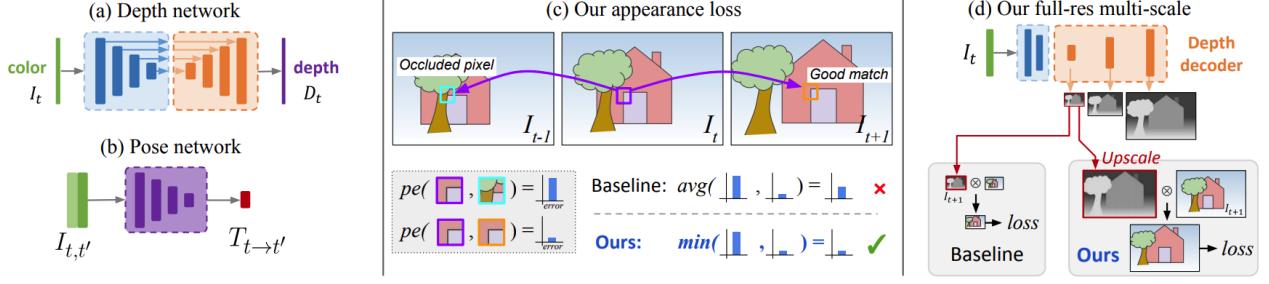


Figure 2.1: Architecture overview of Monodepth2, reproduced from Godard et al. (2019) [1]. The system consists of a depth network that predicts per-pixel depth from a single RGB image and a pose network that estimates relative camera motion between consecutive frames. During training, depth and pose predictions are used to reconstruct neighboring images using a photometric reconstruction loss.

The Monodepth2 architecture consists of two main components: a depth network and a pose network. The depth network follows an encoder–decoder structure and predicts a dense depth map from a single RGB image, while the pose network estimates relative camera motion between consecutive frames. Using these predictions, neighboring images are reconstructed and compared with the original images through a photometric loss. To improve robustness, Monodepth2 introduces a minimum reprojection loss to handle occlusions and an auto-masking strategy to ignore pixels that violate the rigid scene assumption, such as moving objects or stationary cameras. At inference time, only the depth network is required, enabling efficient depth prediction from a single image. [1]

2.2 ZoeDepth: Metric Monocular Depth Estimation

ZoeDepth is a monocular depth estimation method designed to predict depth directly in metric units from a single RGB image. Unlike many earlier monocular approaches that focus mainly on relative depth, ZoeDepth explicitly aims to recover absolute scale, which is essential for safety-critical applications such as autonomous driving and agricultural machinery. The core idea of ZoeDepth is to combine classification-based depth binning with continuous regression, allowing the model to represent both near and far depth ranges effectively. [2]

By learning adaptive depth bins from data, ZoeDepth can better capture complex depth distributions commonly found in outdoor scenes. This hybrid formulation allows the model to benefit from the stability of discretized depth representations while still producing continuous depth estimates. As a result, ZoeDepth provides a practical compromise between relative-depth methods and fully supervised metric depth estimation. [2]

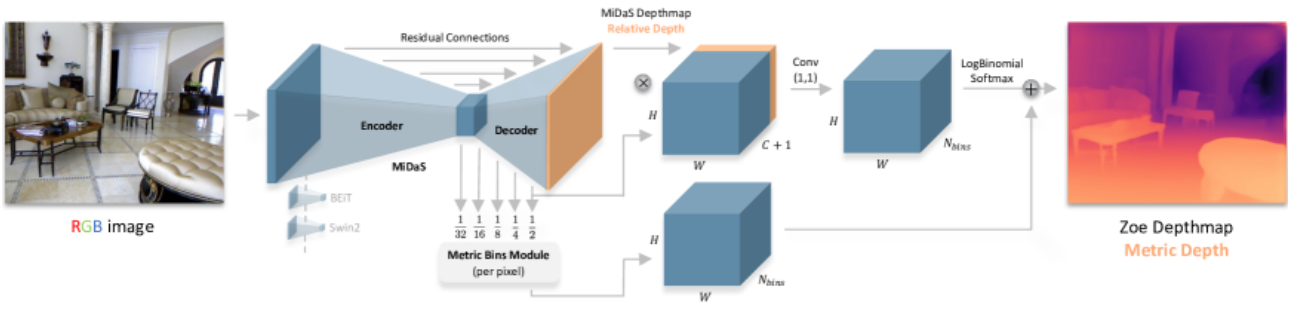


Figure 2.2: Architecture overview of ZoeDepth, reproduced from Bhat et al. (2023) [2]. A shared encoder extracts image features, which are used to estimate adaptive depth bins and corresponding per-pixel depth values, enabling metric depth prediction from a single RGB image.

The ZoeDepth architecture consists of a shared image encoder followed by two prediction branches. One branch estimates a set of adaptive depth bins that represent different depth ranges in the scene, while the other branch predicts per-pixel depth values by regressing over these bins. During training, ZoeDepth is supervised using metric depth data, allowing it to learn absolute scale rather than relative depth only. At inference time, the model produces a dense depth map in metric units from a single RGB image, making it well suited for real-world deployment where accurate distance measurements are required. [2]

2.3 Depth Anything 3: Unified Geometry and Monocular Depth Estimation

Depth Anything 3 (DA3) is a recent visual geometry model that extends monocular depth estimation toward a more general and unified understanding of scene geometry. Instead of relying on task-specific components, DA3 employs a single transformer-based backbone to predict spatially consistent geometry from visual inputs. The model is designed to operate with either single images or multiple views and can predict dense depth maps, camera pose, and related geometric information within a unified framework. [3]

One of the main motivations behind DA3 is architectural simplicity combined with large-scale pretraining. By avoiding specialized depth or pose networks and focusing on a unified depth-ray representation, the model achieves strong generalization across diverse scene types. This makes DA3 particularly relevant for agricultural environments, where lighting conditions, scale, and object appearance can vary significantly. [3]

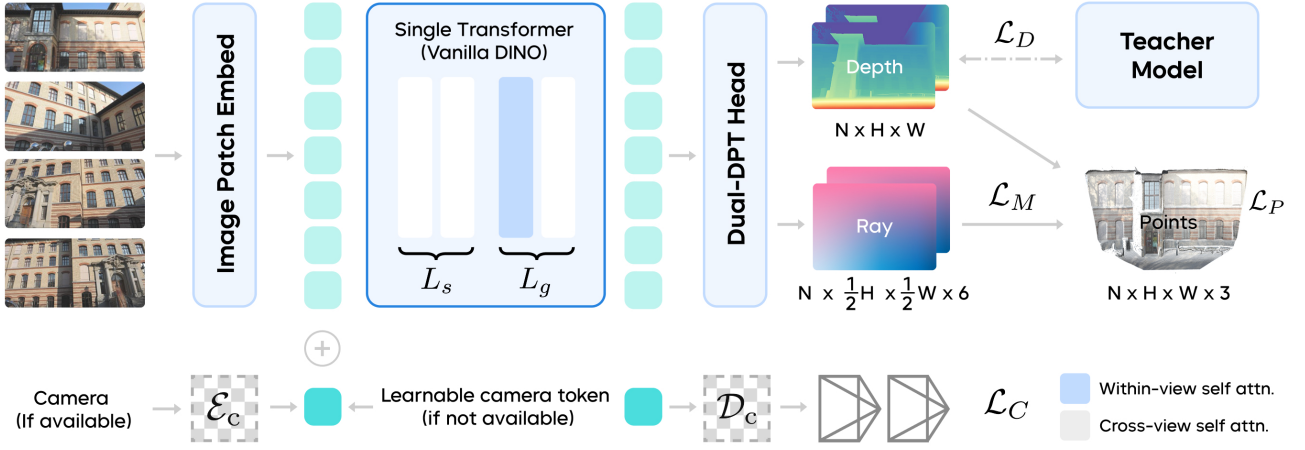


Figure 2.3: Architecture overview of Depth Anything 3, reproduced from Yang et al. (2024) [3]. A single transformer backbone extracts visual features, which are used for unified geometry prediction, including dense depth maps and camera pose.

The DA3 architecture is built around a plain transformer encoder, such as DINOv2, which processes visual input features without requiring separately designed depth or pose networks. From this shared representation, the model predicts a unified set of geometric outputs. For monocular depth estimation, DA3 focuses on producing spatially consistent depth predictions by learning depth rays that encode scene structure. This unified training objective allows the model to handle both monocular and multi-view settings and to achieve strong geometric accuracy while remaining competitive in monocular depth estimation benchmarks. [3]

3 Dataset Preparation

The camera recordings made through Raw LiDAR cannot be directly utilized for monocular depth estimation. Some practical issues need to be addressed, such as fisheye distortion, LiDAR sparsity, and sensor misalignment. The aim is to make the raw recordings of the ROS bag into paired rectified RGB images and LiDAR-based depth maps.

3.1 Data Preparation Pipeline

Figure 3.1 - represents the entire process of data preparation and generation of the depth maps. The raw ROS bag files, including camera images, LiDAR point clouds, and TF data, are converted into ROS2 packages and then filtered to ensure they contain valid LiDAR data. The camera images are decoded and rectified based on the intrinsic and distortion parameters, while the LiDAR point cloud is transformed into the camera coordinate system based on the extrinsic calibration that is available. However, if needed, manual adjustments are performed based on where alignment issues are observed, before the point cloud is projected onto the image plane. Finally, the generated depth maps use a Z-buffering (Section-3.3.4.1) approach and are saved as 16-bit depth maps, along with quality assurance visualizations.

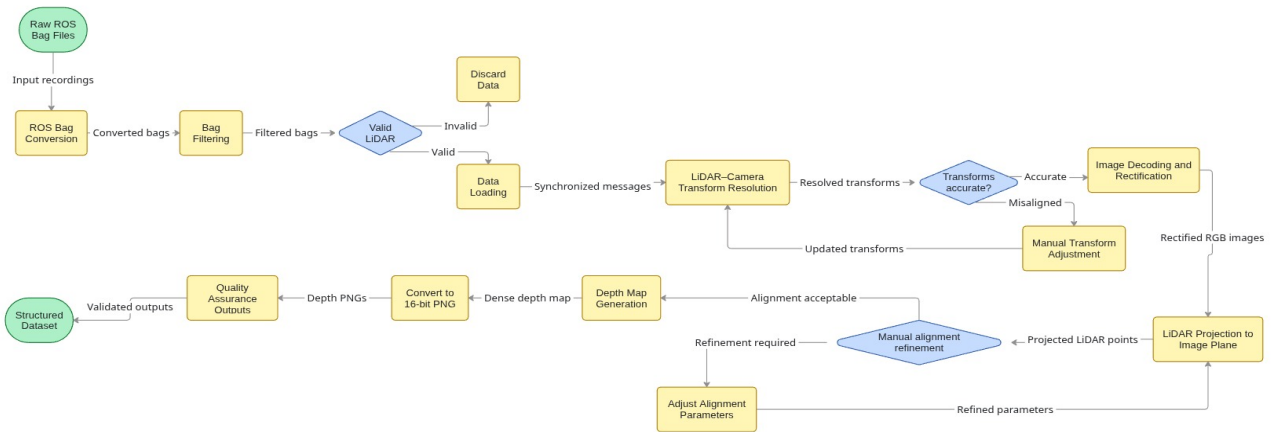


Figure 3.1: Overview of the data preparation and depth map generation pipeline. The pipeline converts raw ROS bag data into rectified RGB images and aligned LiDAR-based depth maps, with optional manual refinement to ensure accurate sensor alignment.

3.2 Dataset Engineering

The dataset used in this research is collected in the form of synchronized image and LiDAR data, saved in ROS bag format. The ROS bag format comprises fisheye RGB images, LiDAR point clouds, and transform data, which describe the relative position of the different

sensors. The data engineering process adopted in this research mainly involves converting the data from one form to another, where the data is in RGB and depth format.

3.2.1 Data Preparation and Dataset Generation

Since raw recordings in ROS format are not usable in direct raw form, not only can they not be utilized due to lens distortion, depth sparseness, and possible misalignment of camera and LIDAR sensors. To address these issues, a dedicated data preparation pipeline was applied to convert the raw recordings into training-ready data.

In order to do this, the first step consisted of transforming all of the existing ROS1 bag files into ROS2-compatible bag folders, complete with database file support, in order to enable robust entry to the relevant sensor messages using contemporary Python-based tools. At this stage, recording filtering is used to exclude recordings that do not possess usable LIDAR point cloud information, with such recordings ignored in favor of those with functional LIDAR support.

Next, the images from the camera were decoded and rectified, taking into consideration known camera calibration parameters. Image rectification removes distortions and maps the original fisheye images into the pinhole camera model. This is important so that the images have similar geometric information, allowing them to work properly in a standard convolutional neural network. The rectified images are stored at a particular resolution to maintain uniform input characteristics throughout the dataset.

For every rectified RGB image, a corresponding LiDAR scan was chosen based on temporal proximity. LiDAR point clouds were transformed into the camera coordinate frame using available extrinsic calibration information. When small residual misalignment was observed, manual refinement of the extrinsic parameters was applied, as described in Section 3.3.6.

3.2.2 Depth Map Generation

Depth maps are generated by projecting the transformed LiDAR points onto the rectified image plane. A Z-buffering strategy is used so that, for each pixel, only the closest LiDAR point along the camera viewing ray contributes to the depth value. This approach ensures that occluded or farther points do not overwrite nearer surfaces.

Depth values are stored as 16-bit PNG images representing metric depth in millimeters as in Figure-3.2. Pixels without valid LiDAR measurements remain zero-valued and are explicitly marked as invalid. These invalid pixels are excluded during model training and evaluation to ensure that learning is guided only by reliable depth measurements.

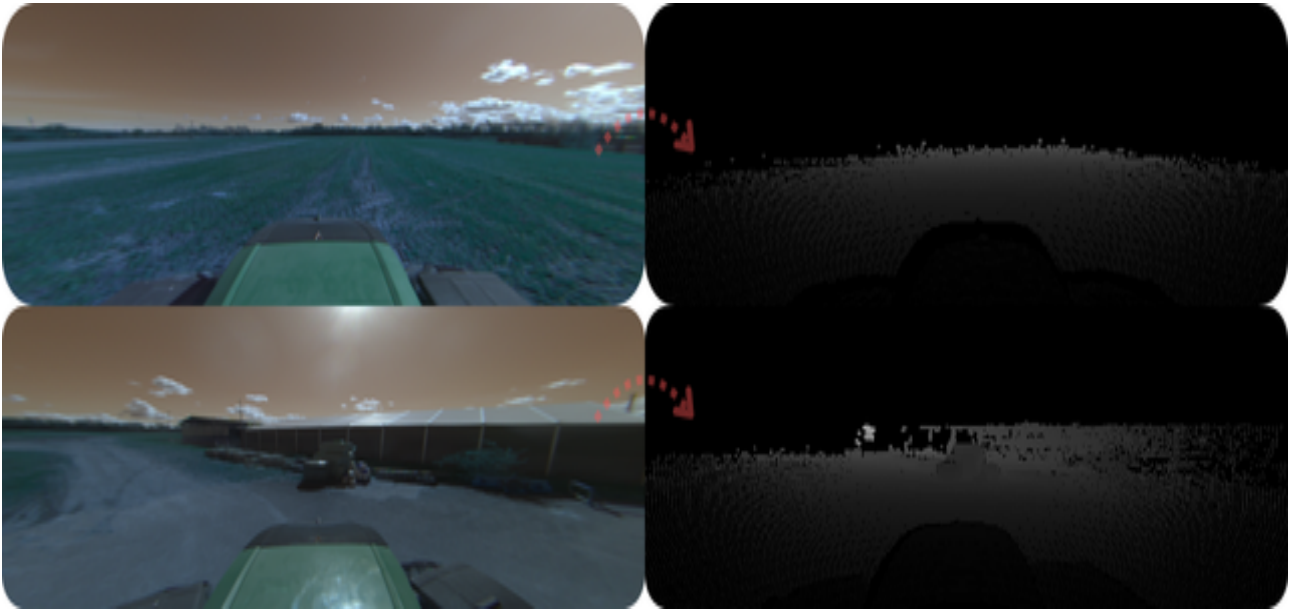


Figure 3.2: Rectified images and their sparse depth maps (in mm)

3.2.3 Depth Smoothing

To increase training stability, an additional option for a depth smoothing strategy is employed as showed Figure 3.3. This process fills small gaps near valid LiDAR measurements using distance-limited interpolation and edge-preserving filtering. Importantly, original LiDAR depth values are preserved, and a validity mask is maintained to distinguish measured depth from filled values.

This design makes it possible to conduct controlled experiments through the use of either "sparse LiDAR depth" or "smoothed depth" based on a common sensor data set.

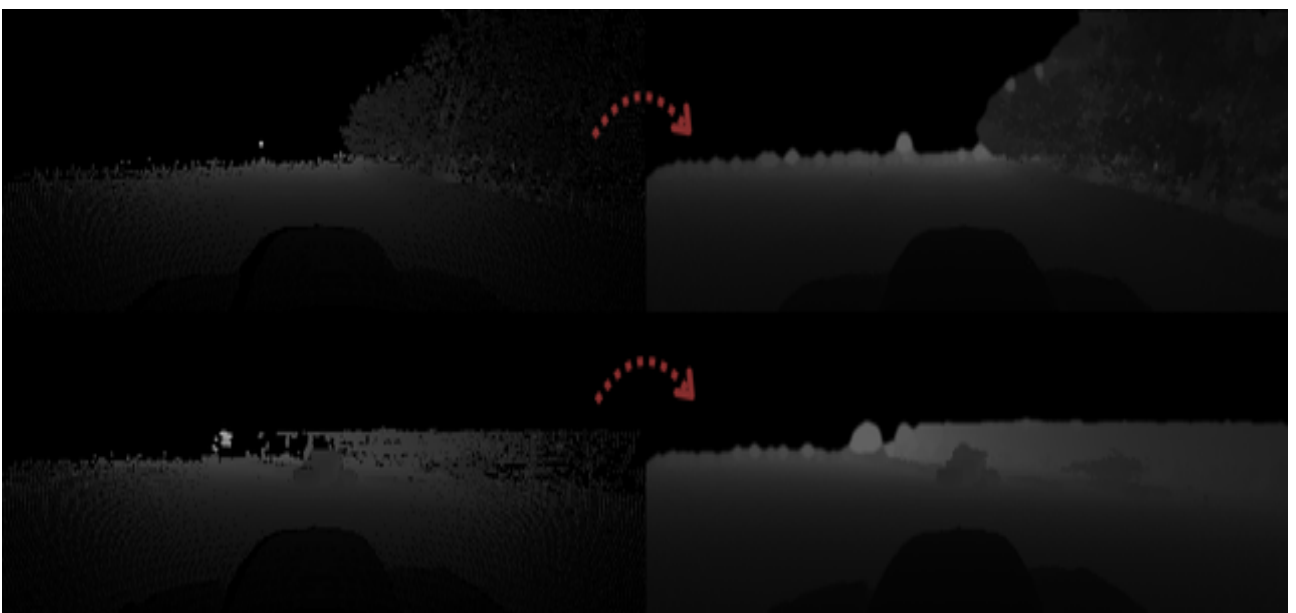


Figure 3.3: Sparse depth maps and their smooth depth maps (in mm)

3.2.4 Dataset Splitting

The final dataset is split at the bag level into training, validation, and test sets. All frames originating from the same ROS bag are assigned to the same split to avoid temporal correlation and scene-level data leakage. There is a predefined set of test bags, which are held completely separate from the training and validation processes.

3.3 LiDAR–Camera Projection

This section describes how three-dimensional LiDAR point clouds are projected onto the two-dimensional camera image plane to generate depth maps aligned with rectified RGB images. This projection is based on geometric transformations considering the camera calibration and LiDAR-camera extrinsics.

3.3.1 LiDAR Point Representation

Each LiDAR scan is represented as a set of three-dimensional points in the LiDAR coordinate frame:

$$\mathbf{P}_{\text{LiDAR}} = \begin{bmatrix} X_L & Y_L & Z_L & 1 \end{bmatrix}^T \quad (3.1)$$

where X_L , Y_L , and Z_L denote the spatial coordinates of a LiDAR point expressed in meters. Homogeneous coordinates are used to enable matrix-based transformations.

3.3.2 Transformation from LiDAR Frame to Camera Frame

To align LiDAR points with the camera image, each point is transformed into the camera coordinate frame using a rigid-body transformation:

$$\mathbf{P}_{\text{Cam}} = \mathbf{T}_{\text{Cam} \leftarrow \text{LiDAR}} \cdot \mathbf{P}_{\text{LiDAR}} \quad (3.2)$$

where $\mathbf{T}_{\text{Cam} \leftarrow \text{LiDAR}} \in \mathbb{R}^{4 \times 4}$ is the extrinsic transformation matrix composed of a rotation matrix $\mathbf{R}$ and a translation vector $\mathbf{t}$:

$$\mathbf{T}_{\text{Cam} \leftarrow \text{LiDAR}} = \begin{bmatrix} \mathbf{R} & \mathbf{t} \\ \mathbf{0} & 1 \end{bmatrix} \quad (3.3)$$

Only points with positive depth in the camera frame ($Z_C > 0$) are retained for further processing.

3.3.3 Projection onto the Image Plane

Once transformed into the camera coordinate frame, LiDAR points are projected onto the two-dimensional image plane using the pinhole camera model:

$$u = f_x \frac{X_C}{Z_C} + c_x \quad (3.4)$$

$$v = f_y \frac{Y_C}{Z_C} + c_y \quad (3.5)$$

where (f_x, f_y) are the focal lengths and (c_x, c_y) denote the principal point coordinates derived from the camera intrinsic matrix.

Projected pixel coordinates are discretized and checked against image boundaries to ensure that only valid pixels contribute to the depth map.

3.3.4 Depth Assignment and Z-Buffering

For each valid projected point, the depth value is defined as the distance along the camera optical axis:

$$D(u, v) = Z_C \quad (3.6)$$

When multiple LiDAR points project to the same pixel, a Z-buffering strategy is applied:

$$D(u, v) = \min(Z_C) \quad (3.7)$$

This ensures that the closest surface along each viewing ray is preserved.

3.3.4.1 Z-buffering explanation

Where it projects points onto an image, as can be expected, multiple 3D points project to a single point on an image. This mostly occurs when an object, which is far away, is obscured from view by another, closer object pointing in the same direction as the viewing position.

Z-buffering solves this by keeping only the *nearest* point to the camera for each pixel. For example, imagine a fence standing in front of a tree. LiDAR returns points from both the fence and the tree, but when projected onto the image they may overlap at the same pixel. Z-buffering compares their depths and keeps the fence point (smaller Z_C), while discarding the tree point behind it.

This ensures that each pixel in the depth map represents the depth of the surface that is actually visible from the camera viewpoint, rather than a surface hidden behind another object.

3.3.5 Depth Map Representation

The resulting depth map is stored as a single-channel image with depth values encoded in millimeters using 16-bit unsigned integers:

$$D_{\text{mm}} = \text{clip}(D_{\text{m}} \times 1000) \quad (3.8)$$

Pixels without valid LiDAR measurements remain zero-valued and are marked as invalid using a corresponding depth validity mask.

3.3.6 Manual Extrinsic Refinement

Although LiDAR–camera extrinsic transformations are available through TF data, small residual misalignments were found in practice. Such misalignments are common in real-world industrial datasets and can result from calibration drift, mounting tolerances, or imperfect synchronization. Even minor extrinsic errors can lead to visible misprojection of LiDAR points onto the image plane.

To address this issue, a controlled manual refinement step was introduced during depth map generation. For each recording, small corrective adjustments were allowed on top of the existing extrinsic transformation. These adjustments consist of minor changes to yaw, pitch, roll, and translation components. When valid TF data was available, the refinement was applied as a delta correction to the TF-derived transformation. In cases where TF information was missing or unreliable, the refined transformation was used directly.

The refinement process is guided by visual inspection of projected LiDAR points over rectified RGB images. Adjustments are kept minimal and are applied only when clear misalignment is observed. Once satisfactory alignment is achieved, the refined parameters are stored and reused automatically for subsequent processing of the same recording. This refinement step improves geometric alignment while preserving the underlying depth generation logic, making the resulting dataset well-suited for real-world agricultural scenarios.

3.4 Dataset Construction and Variants

This section describes how the prepared RGB images and LiDAR-based depth maps are organized in datasets for training, validation, and testing. The dataset preparing process is designed to ensure correct RGB–depth pairing, prevent data leakage across splits, and support multiple ground-truth representations derived from the same sensor data.

3.4.1 Dataset Structure and Sample Definition

After depth map generation, the dataset is organized into a structured directory format consisting of rectified RGB images and corresponding depth maps. Each dataset sample is defined as a tuple:

$$\mathcal{S}_i = (I_i, D_i, M_i) \quad (3.9)$$

where I_i is a rectified RGB image, D_i is the associated LiDAR-derived depth map, and M_i is a binary validity mask indicating pixels with reliable depth values.

RGB images and depth maps are paired using a shared frame index extracted from their filenames. For each RGB image, the nearest depth map with the same frame index is selected, ensuring temporal consistency between the two modalities.

3.4.2 Bag-Level Dataset Splitting

To avoid temporal correlation and scene overlap between training and evaluation data, dataset splitting is performed at the bag level rather than at the individual frame level. All frames originating from the same ROS bag are assigned to the same dataset split.

The dataset is divided into three subsets:

- **Training set**, used for model optimization.
- **Validation set**, used for performance monitoring and model selection.
- **Test set**, consisting of a fixed set of bags reserved exclusively for final evaluation.

A fixed list of test bags is defined to ensure reproducibility across experiments. The remaining bags are randomly shuffled and split into training and validation sets using a predefined validation fraction and random seed.

3.4.3 Sparse Ground-Truth Dataset

The sparse ground-truth dataset directly uses the LiDAR-projected depth maps generated during data preparation as shown in Figure-3.2. In this variant, depth values are available only at pixel locations where LiDAR points were successfully projected onto the image plane.

The validity mask for the sparse dataset is defined as:

$$M_{\text{sparse}}(u, v) = \begin{cases} 1, & \text{if } D(u, v) > 0 \\ 0, & \text{otherwise} \end{cases} \quad (3.10)$$

Pixels without LiDAR coverage remain invalid and are excluded from loss computation during training and evaluation. This dataset variant preserves the original sparsity and measurement characteristics of the LiDAR sensor.

3.4.4 Smooth Ground-Truth Dataset

In addition to the sparse dataset, a smoothed ground-truth variant is constructed to provide a denser supervision signal as shown in Figure-3.3. This variant applies limited depth filling around valid LiDAR pixels while preserving the original depth values at measured locations.

Depth filling is restricted to pixels within a predefined spatial distance from valid LiDAR measurements. Let M_{GT} denote the original sparse validity mask. A fill mask M_{fill} is computed by identifying pixels whose distance to the nearest valid pixel is below a specified threshold. The resulting validity mask is defined as:

$$M_{\text{smooth}} = M_{\text{GT}} \cup M_{\text{fill}} \quad (3.11)$$

Filled depth values are generated using local interpolation and optional edge-preserving filtering. Original LiDAR depth values are kept unchanged to ensure that measured data remains authoritative.

3.4.5 Depth Value Representation and Scaling

All depth maps are stored as single-channel images using 16-bit unsigned integer encoding, representing depth in millimeters. During dataset loading, depth values are converted to meters:

$$D_{\text{m}} = \frac{D_{\text{mm}}}{1000} \quad (3.12)$$

Depth values are clipped to a predefined minimum and maximum range to remove invalid or extreme values. The corresponding validity masks are resized using nearest-neighbor interpolation to match the network input resolution.

3.4.6 Dataset Usage in Training and Evaluation

Both sparse and smooth dataset variants share the same RGB images and bag-level splits. Only the depth map and validity mask used for training and evaluation differ between dataset variants.

During model optimization, loss functions are only applied to pixels which the corresponding mask marked as valid. This ensures that learning is guided only by reliable depth information, regardless of the chosen ground-truth variant.

Since the pipeline generates several dataset variants from the same sensor recordings, it allows for the controlled comparison of LiDAR sparse versus smooth supervision. The data splits and evaluation protocols are the same.

4 Model Training and Evaluation

In this chapter, the overall use of the prepared dataset in relation to the training and evaluation of monocular depth estimation algorithms is explained, including the training methodology, loss functions, the selected models, and the evaluation metrics used for benchmarking.

4.1 Model Training Strategy

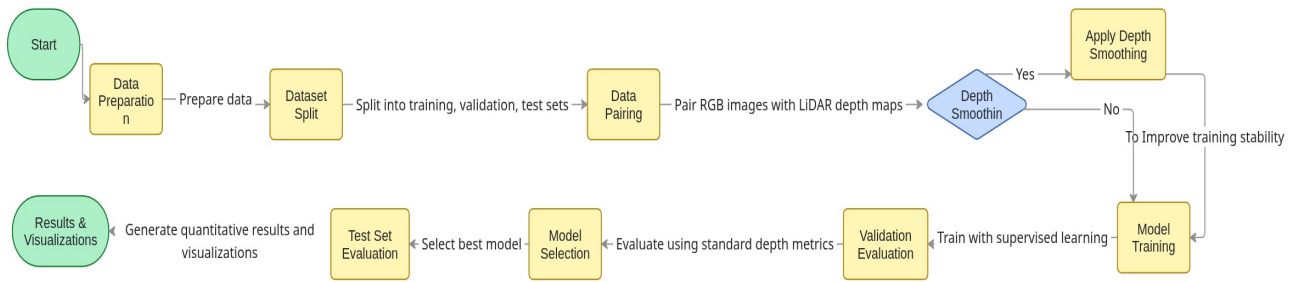


Figure 4.1: Training and evaluation workflow for monocular depth estimation. The pipeline includes dataset splitting, RGB-depth pairing, optional depth smoothing, supervised model training, validation-based model selection, and final evaluation on a fixed test set.

This section explains how the monocular depth estimation models are trained using LiDAR-based depth supervision. The goal of the training process is to learn a mapping from a single RGB image to a dense depth map while using sparse LiDAR depth measurements as reliable ground truth.

The training strategy is designed to handle sparse depth data, ensure stable optimization, and preserve meaningful depth structure in the predicted output.

4.1.1 Training Overview

During training, a single RGB image is provided as input to the depth estimation model. The model predicts a dense depth map with one depth value per pixel. This predicted depth is then compared against the LiDAR-based ground-truth depth map.

Training is performed only on the training split of the dataset. The validation split is used to monitor performance after each training epoch, and the test split is never used during training. All models are initialized from pretrained checkpoints and fine-tuned on the AGCO dataset.

4.1.2 Depth Prediction

Let I denote an input RGB image. The depth estimation model is represented as a function $f(\cdot)$, which produces a predicted depth map:

$$D_{\text{pred}} = f(I) \quad (4.1)$$

The predicted depth map D_{pred} has one depth value for each pixel and is expressed in metric units (meters).

If the spatial resolution of the predicted depth map differs from that of the ground-truth depth map, the ground-truth depth and corresponding validity mask are resized to match the prediction resolution using nearest-neighbor interpolation. This avoids introducing artificial depth values.

4.1.3 Valid Pixel Masking

LiDAR sensors do not provide depth values for all image pixels. As a result, the ground-truth depth map is sparse. To ensure that the model is trained only on reliable depth measurements, a binary validity mask is used.

Let D_{gt} be the ground-truth depth map and M be the validity mask:

$$M(u, v) = \begin{cases} 1, & \text{if } D_{\text{gt}}(u, v) \in [D_{\text{min}}, D_{\text{max}}] \\ 0, & \text{otherwise} \end{cases} \quad (4.2)$$

Only pixels where $M(u, v) = 1$ contribute to the loss. Pixels without valid depth measurements are ignored.

This masking step is essential because it prevents invalid or missing LiDAR values from corrupting the training signal.

4.2 Loss Functions

The primary supervision signal for training is based on masked depth losses computed over valid pixels. In addition, a smoothness term is used to encourage spatial consistency in the predicted depth maps.

4.2.1 Masked Absolute Depth Loss

The primary supervision signal is provided by a masked L1 loss computed over valid pixels:

$$\mathcal{L}_{L1} = \frac{1}{N} \sum_{(u,v)} M(u,v) \cdot |D_{\text{pred}}(u,v) - D_{\text{gt}}(u,v)| \quad (4.3)$$

where $N = \sum_{u,v} M(u,v)$ is the total number of valid pixels.

Why this loss is used:

- The absolute difference directly measures depth error in meters.
- Using L1 loss makes the training less sensitive to outliers compared to squared loss.
- Dividing by N ensures that the loss is independent of the number of valid pixels.

This loss encourages accurate metric depth prediction at locations where LiDAR depth is available.

4.2.2 Masked Logarithmic Depth Loss

In addition to the absolute depth loss, a logarithmic depth loss is used:

$$\mathcal{L}_{\log} = \frac{1}{N} \sum_{(u,v)} M(u,v) \cdot |\log D_{\text{pred}}(u,v) - \log D_{\text{gt}}(u,v)| \quad (4.4)$$

Why logarithmic loss is used:

- Logarithmic depth reduces sensitivity to absolute scale differences.
- It emphasizes relative depth structure rather than raw distance.
- Errors at near and far depths are treated more evenly.

This loss is especially useful for monocular depth estimation, where scale ambiguity is common.

4.2.3 Edge-Aware Smoothness Regularization

To encourage spatial consistency in the predicted depth map, an edge-aware smoothness loss is applied. This loss penalizes large depth differences between neighboring pixels, while allowing depth discontinuities at object boundaries.

The smoothness term is defined as:

$$\mathcal{L}_{\text{smooth}} = |\nabla D_{\text{pred}}| \cdot e^{-|\nabla I|} \quad (4.5)$$

Explanation of terms:

- ∇D_{pred} represents depth gradients.
- ∇I represents image intensity gradients.
- The exponential weighting reduces smoothing near strong image edges.

This regularization helps produce cleaner and more realistic depth maps without over-smoothing object boundaries.

4.2.4 Total Training Loss

The final training loss is a weighted combination of the individual loss terms:

$$\mathcal{L}_{\text{total}} = \lambda_1 \mathcal{L}_{\text{L1}} + \lambda_2 \mathcal{L}_{\text{log}} + \lambda_3 \mathcal{L}_{\text{smooth}} \quad (4.6)$$

Why weighted combination is used:

- Each loss term addresses a different aspect of depth quality.
- Weighting allows controlled contribution of each term.
- The weights remain fixed during training.

If a training sample contains no valid depth pixels, loss computation for that sample is skipped to avoid numerical instability.

4.2.5 Training Procedure

Training is performed for a fixed number of epochs using mini-batch optimization. After each epoch, the model is evaluated on the validation set using standard depth estimation metrics. The model achieving the best validation performance is saved for final evaluation. This training strategy ensures that learning is driven by reliable LiDAR depth measurements while maintaining smooth and physically meaningful depth predictions.

4.2.6 Models Used in Experiments

This chapter evaluates three monocular depth estimation models: Monodepth2 [1], ZoeDepth [2], and Depth Anything v3 [3]. These models were reviewed in detail in Chapter-2. They are treated as established depth prediction backbones and compared under a unified training and evaluation framework.

All models are fine-tuned using the same LiDAR-supervised dataset, identical data splits, and consistent evaluation metrics. This controlled setup ensures that observed performance differences can be attributed to model design and supervision strategy, rather than dataset or training inconsistencies.

Monodepth2 [1] represents a self-supervised baseline that primarily learns relative depth and relies on scale recovery. ZoeDepth [2] introduces a hybrid formulation that combines relative depth with metric depth regression through adaptive depth bins. Depth Anything v3 [3] reflects a more recent large-scale pretrained model designed for strong metric depth prediction and generalization. Together, these models span a representative range of monocular depth estimation paradigms, enabling a meaningful comparative analysis on the agricultural dataset.

4.3 Evaluation Metrics

This section describes the evaluation metrics used to measure the performance of monocular depth estimation models. All metrics follow the standard protocol introduced in Monodepth2 [1] and are widely used for benchmarking depth estimation models.

The predicted depth maps are compared against LiDAR-based ground-truth depth maps. Since LiDAR depth is sparse, evaluation is performed only on pixels where valid ground-truth depth measurements are available.

4.3.1 Evaluation Setup and Notation

Let $D_{\text{gt}}(u, v)$ denote the ground-truth depth at pixel location (u, v) , and let $D_{\text{pred}}(u, v)$ denote the corresponding predicted depth. Both values are expressed in meters.

A validity mask $M(u, v)$ is defined to indicate reliable depth measurements:

$$M(u, v) = \begin{cases} 1, & \text{if } D_{\text{gt}}(u, v) \in [D_{\text{min}}, D_{\text{max}}] \\ 0, & \text{otherwise} \end{cases} \quad (4.7)$$

Only pixels where $M(u, v) = 1$ are used for evaluation. The total number of valid pixels is,

$$N = \sum_{u,v} M(u, v) \quad (4.8)$$

Dividing by N in all metrics ensures that errors are averaged per pixel, making the metrics independent of image resolution or dataset size.

4.3.2 Absolute Relative Error (Abs Rel)

The absolute relative error measures the average relative difference between predicted and ground-truth depth:

$$\text{AbsRel} = \frac{1}{N} \sum_{(u,v)} \frac{|D_{\text{pred}}(u, v) - D_{\text{gt}}(u, v)|}{D_{\text{gt}}(u, v)} \quad (4.9)$$

Why this metric is used:

- The absolute difference $|D_{\text{pred}} - D_{\text{gt}}|$ measures raw depth error.
- Dividing by D_{gt} normalizes the error, making mistakes at different depths comparable.
- This is important because an error of 1 meter is more serious for nearby objects than for distant ones.

Lower values indicate better performance.

4.3.3 Squared Relative Error (Sq Rel)

The squared relative error increases the penalty for large depth errors:

$$\text{SqRel} = \frac{1}{N} \sum_{(u,v)} \frac{(D_{\text{pred}}(u, v) - D_{\text{gt}}(u, v))^2}{D_{\text{gt}}(u, v)} \quad (4.10)$$

Why squaring is used:

- Squaring amplifies large errors more than small ones.
- This metric is sensitive to severe depth failures, such as large overestimation or underestimation.
- Division by ground-truth depth again provides scale awareness.

Lower values indicate fewer large depth errors.

4.3.4 Root Mean Squared Error (RMSE)

RMSE measures the average magnitude of depth errors in absolute metric units:

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{(u,v)} (D_{\text{pred}}(u, v) - D_{\text{gt}}(u, v))^2} \quad (4.11)$$

Why square and root are used:

- Squaring ensures that positive and negative errors do not cancel each other.
- The square root brings the result back to the original unit (meters).
- RMSE strongly penalizes large absolute errors.

Lower RMSE values indicate more accurate metric depth estimation.

4.3.5 Logarithmic RMSE (RMSE Log)

RMSE Log evaluates error in logarithmic depth space:

$$\text{RMSE}_{\log} = \sqrt{\frac{1}{N} \sum_{(u,v)} (\log D_{\text{pred}}(u, v) - \log D_{\text{gt}}(u, v))^2} \quad (4.12)$$

Why logarithms are used:

- Logarithmic depth reduces sensitivity to absolute scale.
- Relative depth structure is emphasized rather than exact distances.
- Errors at near and far depths are treated more evenly.

This metric is particularly useful for monocular models, where scale ambiguity is common.

4.3.6 Threshold Accuracy Metrics ($\delta_1, \delta_2, \delta_3$)

Threshold accuracy metrics measure the proportion of pixels where the predicted depth is within a multiplicative tolerance of the ground truth:

$$\delta(u, v) = \max \left(\frac{D_{\text{gt}}(u, v)}{D_{\text{pred}}(u, v)}, \frac{D_{\text{pred}}(u, v)}{D_{\text{gt}}(u, v)} \right) \quad (4.13)$$

The prediction is considered accurate if $\delta(u, v)$ is below a predefined threshold:

$$\delta_1 : \delta < 1.25 \tag{4.14}$$

$$\delta_2 : \delta < 1.25^2 \tag{4.15}$$

$$\delta_3 : \delta < 1.25^3 \tag{4.16}$$

Why this ratio is used:

- The ratio treats overestimation and underestimation symmetrically.
- Thresholds reflect increasing tolerance levels.
- These metrics are easy to interpret as percentages of correct predictions.

Higher values indicate better performance.

5 Experiments

This chapter presents the experimental design and benchmarking results for monocular depth estimation on the AGCO dataset. It first describes the experimental setup, dataset splits, ground-truth variants, training configuration, and evaluation protocol. It then reports training behaviour (loss and validation trends), followed by quantitative testset results under multiple evaluation settings, and finally a qualitative model comparison with discussion and error analysis.

5.1 Experimental Setup

The experimental setup describes the experimental configuration used in this thesis. It documents the models, datasets, training settings, hardware, and software environments under which all experiments were conducted. The purpose is to ensure reproducibility and to clearly define the conditions under which results were obtained.

Baseline Models

Three monocular depth estimation models are evaluated:

- Monodepth2
- ZoeDepth
- Depth Anything v3 (DA-v3)

All models are initialized from publicly available pretrained checkpoints and are evaluated both in a zero-shot setting and after supervised finetuning on the AGCO dataset.

5.1.1 Dataset Usage and Splits

The AGCO dataset consists of rectified RGB images and corresponding LiDAR-derived depth maps, organized into sequences (bags). To avoid temporal overlap between training and testing data, a bag-level split strategy is used.

A fixed set of seven bags is reserved as the test set and remains unchanged across all experiments. The remaining bags are used for training and validation. A validation subset is selected from the non-test bags using a fixed validation bag fraction.

Within each split, frame-level subsampling is applied using configurable train and validation fractions. This allows experiments with different proportions of the available data while keeping the test set fixed.

5.1.2 Ground-Truth Depth Variants

Two types of LiDAR-based ground-truth depth are used during finetuning:

- **Normal ground truth**, where only original LiDAR depth measurements are used.
- **Smooth ground truth**, where sparse LiDAR depth is locally filled and smoothed within a limited spatial and depth range, and the valid depth mask is expanded accordingly.

Both variants are evaluated using the same dataset split and evaluation protocol.

5.1.3 Training Configuration

Supervised finetuning is performed using sparse LiDAR depth as ground truth. Training is conducted for a fixed number of epochs per model:

- Monodepth2 is finetuned for 10 epochs (e10).
- ZoeDepth and Depth Anything v3 are finetuned for 30 epochs (e30).

Batch size is selected per model based on GPU memory constraints. During training, only pixels with valid depth measurements contribute to the loss.

For each run, the model checkpoint with the best validation performance is selected and saved for final evaluation.

5.1.4 Hardware and Software Environment

Different computing environments are used due to model-specific and hardware constraints:

- **Monodepth2:** trained on an NVIDIA Quadro P2000 GPU with 4 GB GPU memory and 32 GB system RAM. The software stack includes Python 3.6.10, PyTorch 0.4.1, torchvision 0.2.1, and OpenCV 3.3.1.
- **ZoeDepth:** trained on an NVIDIA RTX 4060 GPU with 8 GB GPU memory and 16 GB system RAM. The software stack includes Python 3.9.23, PyTorch 1.13.1, torchvision 0.14.1, and OpenCV 4.6.0.
- **Depth Anything v3:** trained on the same RTX 4060 GPU using Python 3.10.19, PyTorch 2.9.1, torchvision 0.24.1, and OpenCV 4.11.0.

5.1.5 Evaluation Protocol

All models are evaluated on the same fixed test set using identical depth evaluation metrics. Metrics are computed only on valid depth pixels within a predefined depth range to ensure, all models are compared under the same evaluation conditions.

5.2 Datasets Used in Experiments

All experiments in this thesis are conducted using a curated agricultural dataset collected from real-world tractor operations. The dataset consists of rectified RGB images and corresponding LiDAR-derived depth maps, which serve as ground truth for supervised training and evaluation. To ensure fair comparison across models, a consistent dataset split strategy is applied for all experiments.

The dataset is divided at the bag level into three parts: training, validation, and test sets. A fixed set of seven bags is reserved exclusively for testing and remains unchanged across all experiments and models. The remaining bags are split into training and validation subsets, after which frame-level subsampling is applied. Experiments are performed using three different fractions of the available data: 20%, 50%, and 80% of the full dataset. This allows analysis of model behavior under limited, medium, and large data regimes.

Each experiment uses two variants of the dataset:

- **Sparse ground truth dataset**, where only original LiDAR measurements are used as valid depth supervision.
- **Smoothed ground truth dataset**, where small gaps near valid LiDAR points are filled to increase depth density and training stability, while preserving original LiDAR values.

Importantly, the number of RGB images remains identical between sparse and smoothed datasets. The difference lies only in the number of valid depth pixels available for supervision.

Dataset Sizes and Splits

Table 5.1 summarizes the final dataset sizes used in the experiments. The same train, validation, and test splits are shared across Monodepth2, ZoeDepth, and Depth Anything v3 to ensure a controlled and fair comparison.

These dataset sizes are identical for all models and for both sparse and smoothed ground truth variants. This ensures that any performance differences observed in later sections are caused by model design and training strategy rather than differences in data quantity.

Table 5.1: Dataset sizes used in experiments for different data fractions.

Dataset Fraction	Train Images	Validation Images	Test Images
20% / 20%	2,465	544	Fixed (7 bags)
50% / 50%	6,163	1,359	Fixed (7 bags)
80% / 80%	9,861	2,174	Fixed (7 bags)

The same test set is used throughout all experiments, enabling direct comparison between zero-shot baseline performance and fine-tuned models across different dataset sizes and ground truth variants.

5.3 Training Experiments

So many experiments has been conducted to get better result on AGCO dataset using finetuned models of Monodepth2, ZoeDepth and Depth Anything-v3 which shows how (i) the supervised training loss decreases and (ii) the validation performance evolves across epochs. The goal is to understand convergence stability, sensitivity to dataset size, and the impact of sparse versus smoothed supervision. Final test-set numbers are reported separately in Section 5.4.

All fine-tuning runs use the same bag-level splitting strategy and the same three data fractions (20%, 50%, and 80%). For each run, the best checkpoint is selected using validation performance (as logged by the training scripts).

5.3.1 Experiment Groups

I organize the training experiments into three groups:

1. **Zero-shot baseline:** evaluation only (no training curves).
2. **Sparse LiDAR-supervised fine-tuning:** supervision uses only original LiDAR-projected valid pixels.
3. **Smooth LiDAR-supervised fine-tuning:** supervision uses the smoothed depth target (denser supervision), where filled values increase the number of supervised pixels.

5.3.2 Baseline (Zero-shot) Runs

In the baseline setting, models are evaluated using pretrained checkpoints without any AGCO-specific fine-tuning. Since no optimization occurs, there are no training or validation curves for this setting. These baseline results serve as the reference point for measuring the impact of LiDAR-supervised fine-tuning and are reported in Section 5.4.

5.3.3 Monodepth2: Convergence and Validation Trends

Monodepth2 is fine-tuned for 10 epochs in all experiments. Across both sparse and smooth supervision settings, the model shows a clear and consistent convergence behaviour: the training loss drops rapidly during the first few epochs and then gradually stabilizes. This indicates that Monodepth2 adapts quickly to the AGCO agricultural domain, even when trained with limited supervision.

Sparse supervision (Monodepth2). Under sparse LiDAR supervision, Monodepth2 demonstrates stable learning and improved validation behaviour as the amount of training data increases. The training loss decreases sharply in early epochs and continues to improve more smoothly afterwards:

- **20%:** the training loss drops from 0.9955 at epoch 1 to 0.3939 by epoch 10. The validation Abs Rel improves steadily, with the best validation behaviour appearing in the later epochs (around epoch 9–10).
- **50%:** the training loss reduces from 0.7843 to 0.3035, showing improved stability and reduced fluctuations in the validation curve compared to the 20% setting.
- **80%:** the loss decreases further from 0.6882 to 0.2758, and the validation trend becomes smoother and more stable, indicating better generalization.

Overall, increasing the dataset fraction mainly improves validation stability rather than changing the convergence speed, suggesting that Monodepth2 is already data-efficient but benefits from more supervision in terms of robustness.

Smooth supervision (Monodepth2). A similar convergence pattern is observed under smooth supervision. The training loss again drops strongly in the first epochs (for example, from 1.0307 to 0.3934 for the 20% setting), followed by gradual stabilization. Compared to sparse supervision, the validation curves under smooth supervision are typically more consistent and less noisy. This is expected, as smooth supervision provides denser pixel-level signals, allowing the model to learn more uniform depth structures across the image.

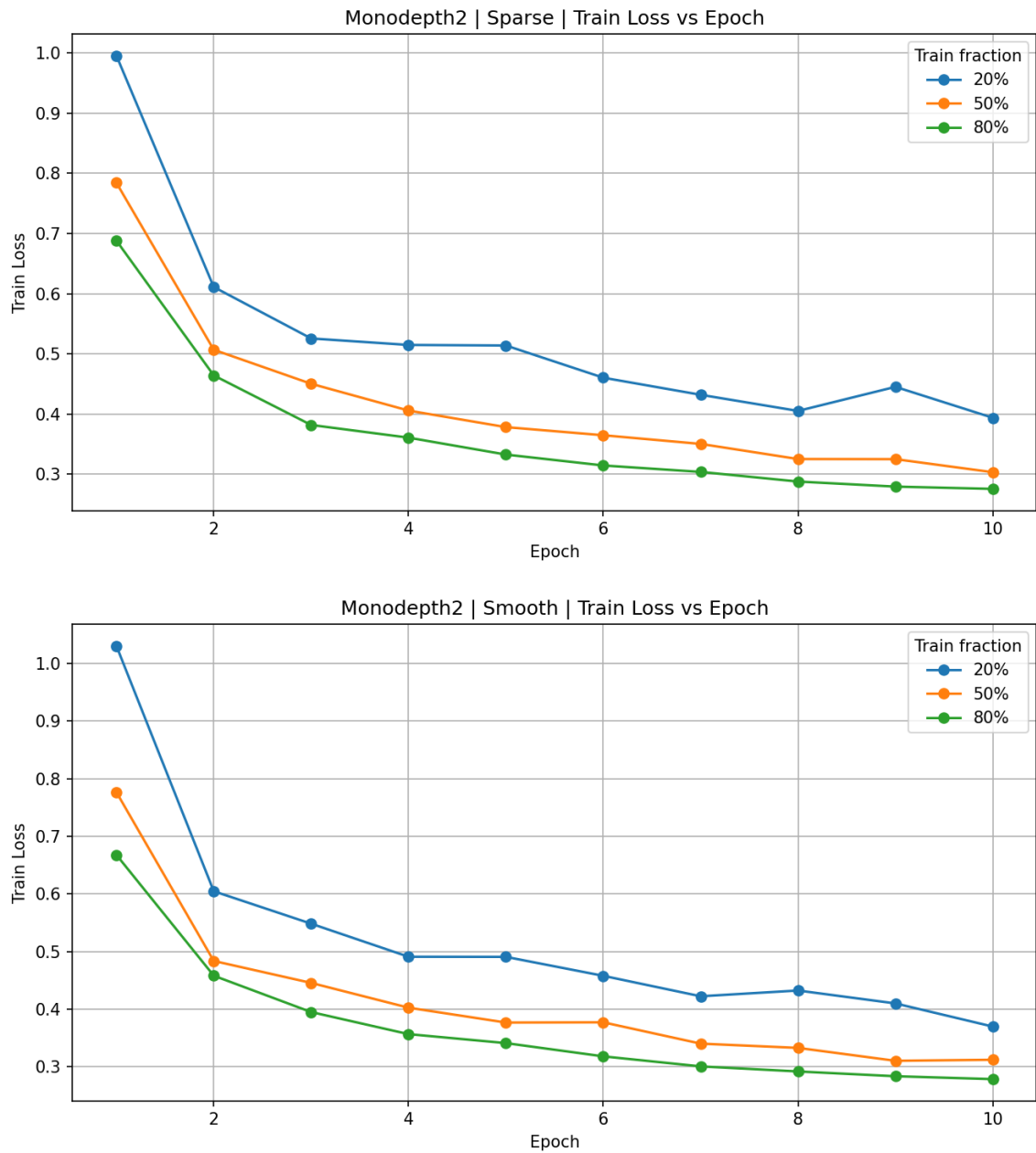


Figure 5.1: Monodepth2 training loss versus epoch for sparse (upper) and smooth (lower) supervision, shown for 20%, 50%, and 80% training fractions.

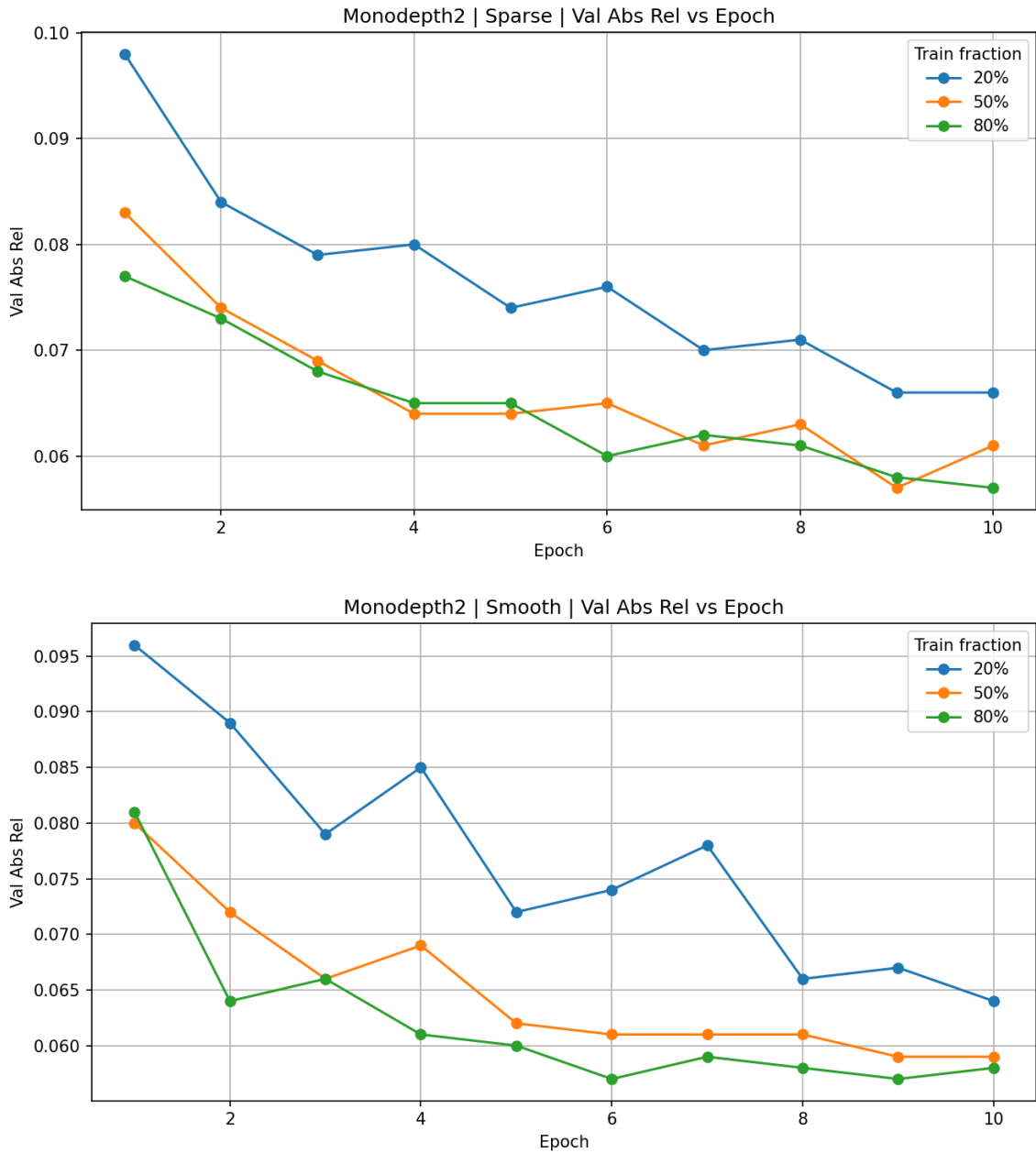


Figure 5.2: Monodepth2 validation Abs Rel versus epoch for sparse (upper) and smooth (lower) supervision across different training fractions.

Training and validation curves. Figure 5.1 shows the training loss versus epoch for the 20%, 50%, and 80% data fractions under both sparse and smooth supervision. The plots clearly illustrate the fast early convergence and the consistent ordering of curves, where larger training fractions achieve lower loss values throughout training.

Figure 5.2 presents the corresponding validation Abs Rel trends. These curves confirm that increasing the training fraction leads to more stable and improved validation performance, particularly in later epochs.

5.3.4 ZoeDepth: Convergence and Validation Trends

ZoeDepth is fine-tuned for 30 epochs per run. Compared to MonoDepth2, the optimization behaviour is noticeably more gradual, with improvements accumulating steadily over a larger number of epochs. This reflects the higher model capacity and the more complex depth-bin formulation used by ZoeDepth, which typically requires longer training to fully adapt to the AGCO domain.

Sparse supervision (ZoeDepth). Under sparse LiDAR supervision, ZoeDepth shows stable and mostly monotonic improvement in both training loss and validation performance. The training loss drops sharply during the first few epochs and then continues to decrease slowly but consistently:

- **20%:** validation Abs Rel improves from 0.188 at epoch 1 to approximately 0.126 at its best point (around epoch 24). The training loss also decreases substantially, from about 2.339 at epoch 1 to values close to 1.1 in later epochs.
- **50%:** the validation Abs Rel further improves, reaching a best value of roughly 0.120 around epoch 23.
- **80%:** the strongest performance is observed for the largest data fraction, with validation Abs Rel improving to around 0.117 (best around epoch 26).

These trends clearly demonstrate that ZoeDepth benefits from increased supervision: larger training fractions lead to lower validation error and smoother convergence, particularly in later epochs.

Smooth supervision (ZoeDepth). In contrast, the smooth supervision setting produces unusually large validation Abs Rel values, typically in the range of 5.4–5.5 across all training fractions. Importantly, this behaviour does *not* indicate a failure to learn. The training loss curves show normal convergence behaviour, and the validation curves remain relatively stable over time.

This discrepancy is instead attributed to the smooth evaluation configuration, where filled or interpolated pixels dominate the relative-error computation. As a result, the Abs Rel metric becomes inflated, even though the predicted depth maps remain visually and structurally reasonable. This interpretation is consistent with the diagnostic experiments presented in Section 5.4, where smooth-trained ZoeDepth models behave normally when evaluation is restricted to LiDAR-valid pixels only.

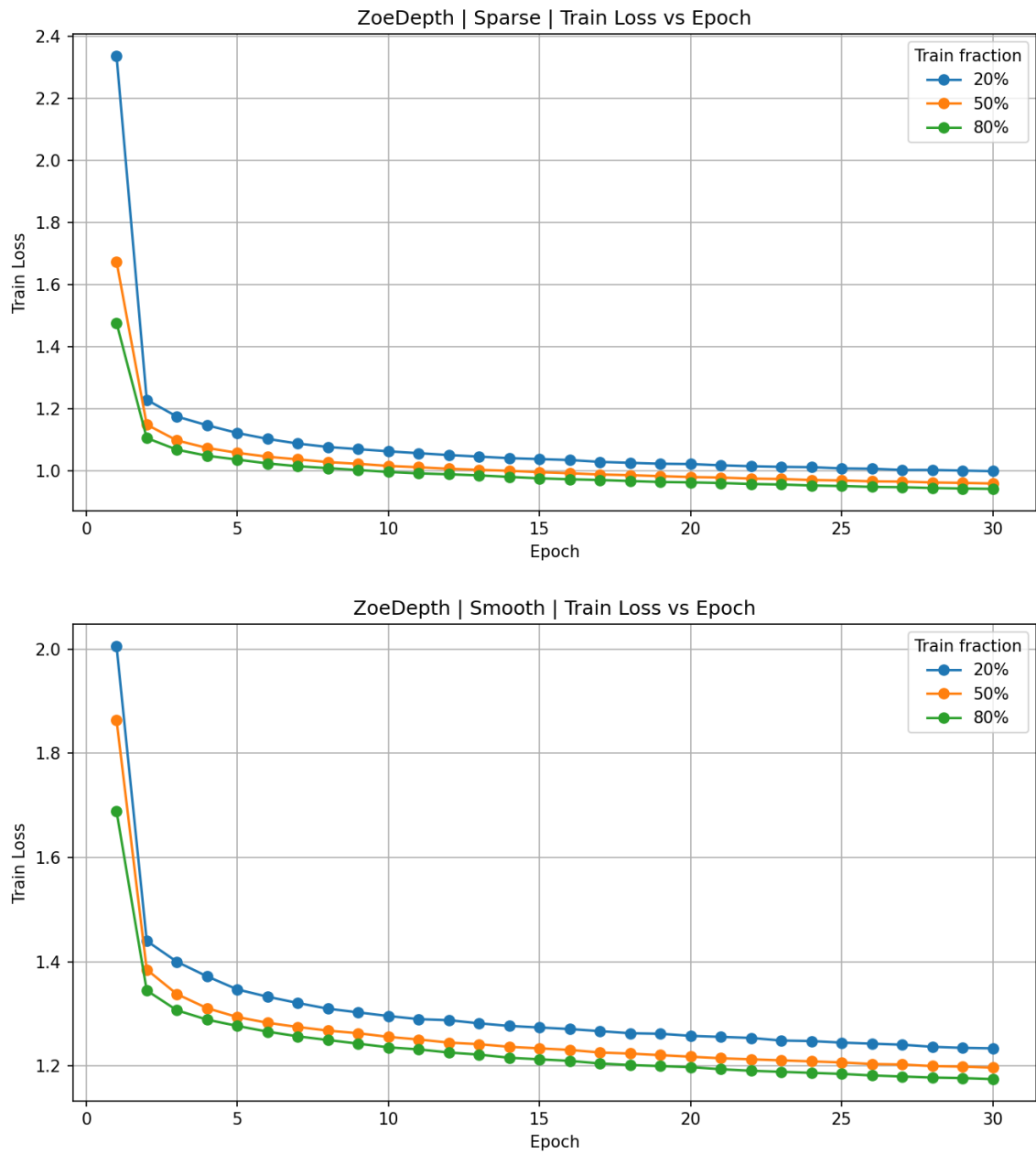


Figure 5.3: ZoeDepth training loss versus epoch for sparse (upper) and smooth (lower) supervision, shown for 20%, 50%, and 80% training fractions.

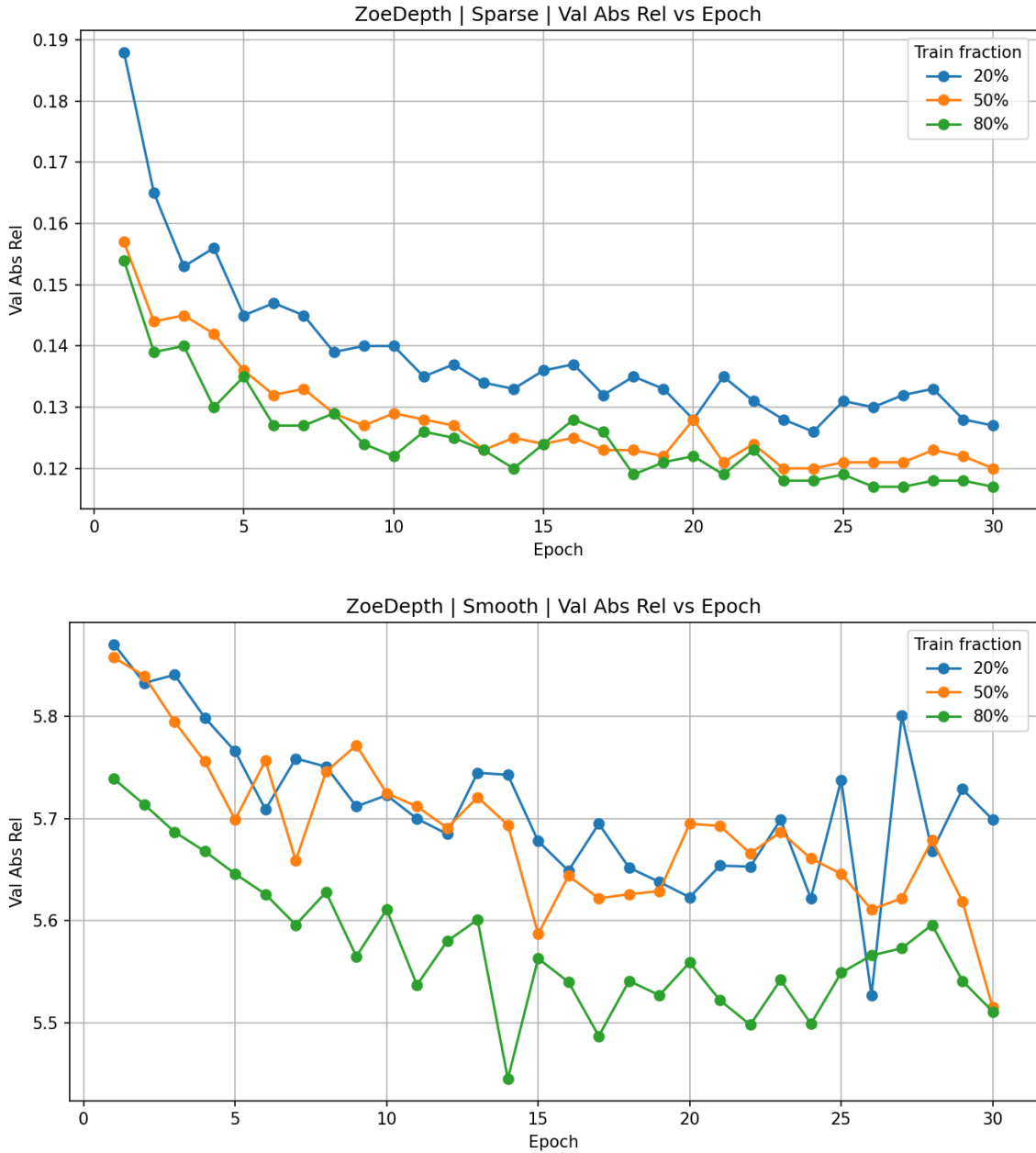


Figure 5.4: ZoeDepth validation Abs Rel versus epoch for sparse (upper) and smooth (lower) supervision across different training fractions.

Training and validation curves. Figure 5.3 shows the training loss versus epoch for the 20%, 50%, and 80% data fractions under sparse and smooth supervision. In both settings, the curves confirm steady convergence over the full 30 epochs, with higher data fractions consistently achieving lower loss values.

Figure 5.4 presents the corresponding validation Abs Rel trends. Under sparse supervision, validation error decreases smoothly and benefits clearly from additional training data. Under smooth supervision, the validation curves remain at a much higher scale, highlighting the strong influence of the evaluation protocol rather than true degradation in model learning.

5.3.5 Depth Anything v3: Convergence and Validation Trends

Depth Anything v3 (DA-v3) is fine-tuned for 30 epochs per run. Compared to MonoDepth2 and ZoeDepth, DA-v3 exhibits the strongest and most consistent convergence behaviour under sparse supervision, and it typically achieves the best validation performance across all dataset fractions. This indicates that DA-v3 is particularly data-efficient and well-suited for adaptation to the AGCO agricultural domain.

Sparse supervision (DA-v3). Under sparse LiDAR supervision, DA-v3 shows rapid early convergence followed by steady refinement throughout training. Both the training loss and validation Abs Rel improve consistently, with clear benefits from increasing the amount of supervision:

- **20%:** validation Abs Rel decreases from 0.158 at epoch 1 into the 0.10 range, with a best logged value around 0.0997 by mid-training.
- **50%:** the best validation Abs Rel improves further to approximately 0.0818, accompanied by smoother and more stable validation curves.
- **80%:** the strongest performance is achieved in this setting, with validation Abs Rel reaching approximately 0.0716, clearly demonstrating the benefit of additional AGCO supervision.

Across all fractions, the training loss decreases sharply in the first few epochs and then gradually stabilizes, indicating fast adaptation combined with reliable long-term optimization. Compared to the other models, DA-v3 consistently attains lower validation error at comparable training stages.

Smooth supervision (DA-v3). Similar to ZoeDepth, the smooth supervision runs of DA-v3 exhibit highly inflated validation Abs Rel values under the smooth evaluation configuration. Despite this, the training loss curves show normal and stable convergence, and the validation curves follow coherent trends over epochs.

This behaviour again suggests that the inflated relative-error values are primarily caused by the evaluation protocol, specifically the treatment of filled or interpolated pixels in the smooth depth maps. As confirmed later by the diagnostic experiments in Section 5.4, DA-v3 models trained with smooth supervision behave normally when evaluation is restricted to LiDAR-valid pixels, indicating that the underlying depth predictions remain meaningful.

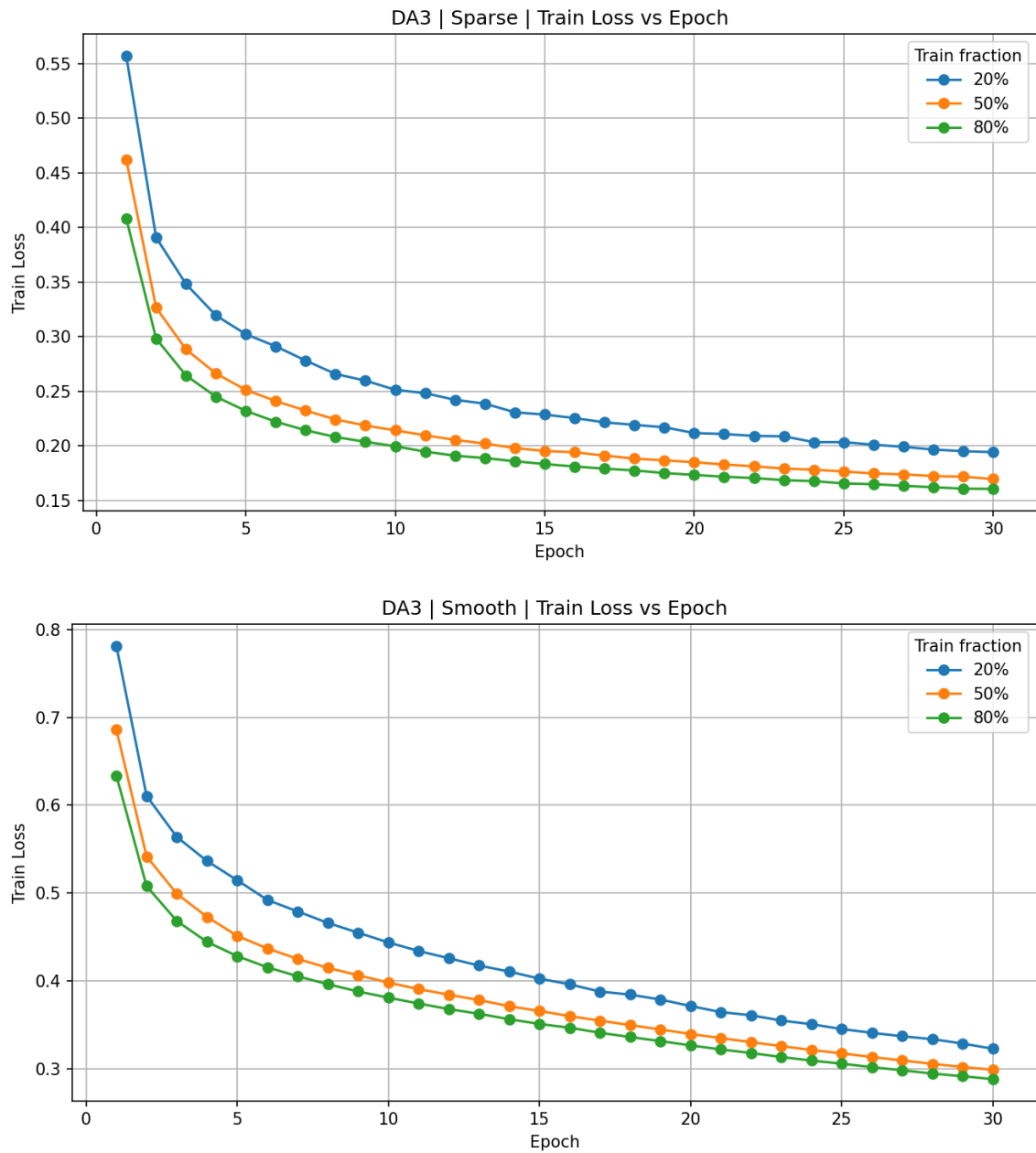


Figure 5.5: DA-v3 training loss versus epoch for sparse (upper) and smooth (lower) supervision, shown for 20%, 50%, and 80% training fractions.

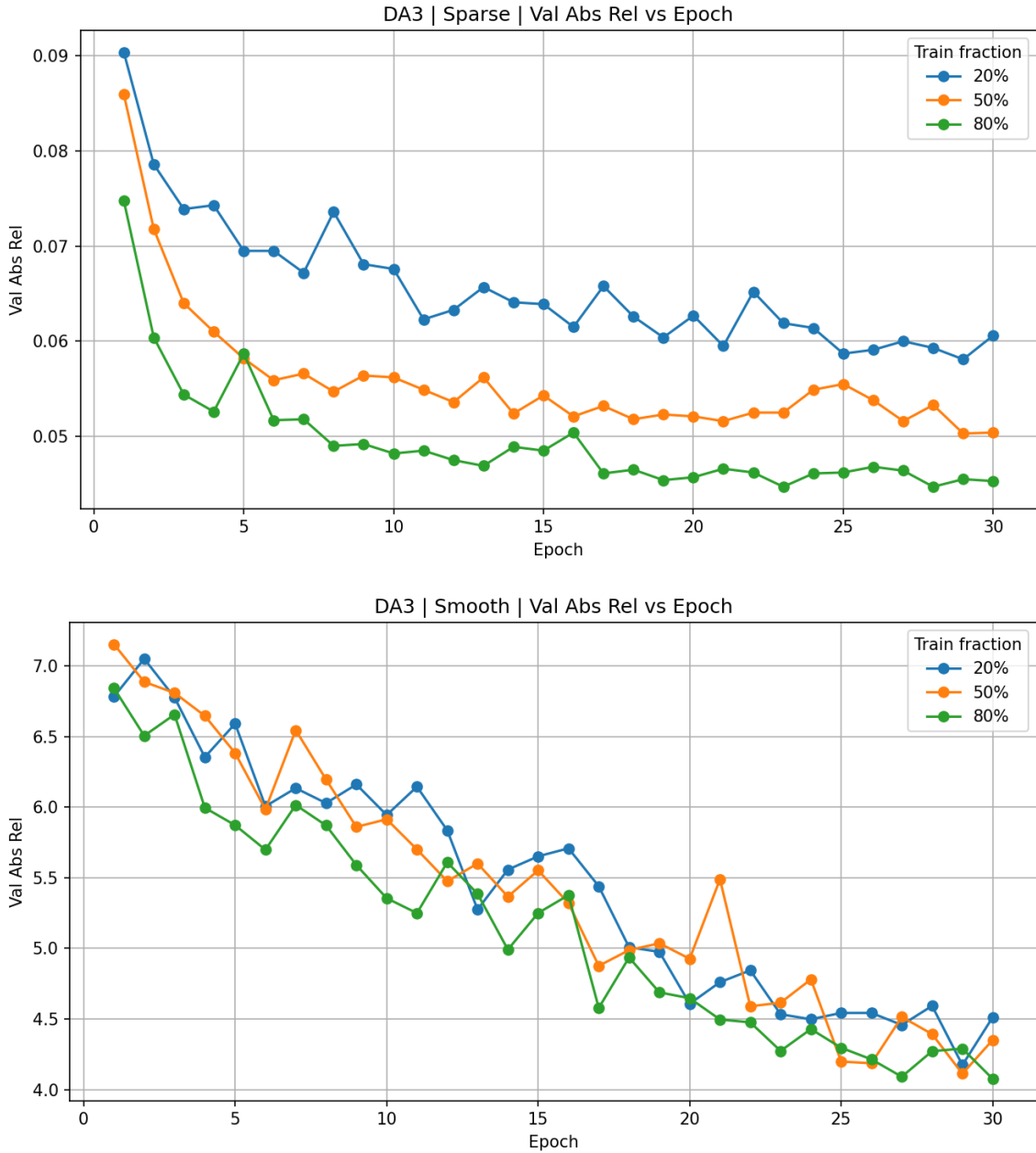


Figure 5.6: DA-v3 validation Abs Rel versus epoch for sparse (upper) and smooth (lower) supervision across different training fractions.

Training and validation curves. Figure 5.5 shows the training loss versus epoch for DA-v3 under sparse and smooth supervision. In both cases, higher training fractions consistently achieve lower loss values, and the overall convergence is smooth and well-behaved across all runs.

Figure 5.6 presents the corresponding validation Abs Rel trends. Under sparse supervision, DA-v3 achieves the lowest validation errors among the three evaluated models, with clear and monotonic improvement as the dataset fraction increases. Under smooth supervision, the validation curves remain at a much higher scale, reinforcing the conclusion that evaluation masking plays a dominant role in the reported error values.

5.3.6 Training Behaviour

Across all evaluated models, sparse fine-tuning consistently leads to improved validation performance as training progresses and as the available training data increases from 20% to 50% and 80%. In all cases, higher data fractions result in lower validation error and more stable convergence behaviour, indicating improved generalization to the AGCO domain. MonoDepth2 exhibits very fast convergence, reaching a stable training and validation regime within 10 epochs. In contrast, ZoeDepth and Depth Anything v3 require longer optimization, showing slower but steady improvements over 30 epochs. This difference reflects the higher model capacity and more complex prediction mechanisms of ZoeDepth and DA-v3, which benefit from extended training.

Under smooth supervision, MonoDepth2 remains stable and behaves similarly to its sparse counterpart. However, for ZoeDepth and DA-v3, the smooth supervision setting produces unusually inflated relative-error validation metrics when evaluated under the smooth evaluation protocol. Importantly, this behaviour does not indicate poor learning; rather, it arises from an interaction between the evaluation metric and the ground-truth variant used for smooth supervision. This effect is therefore treated as an evaluation artefact and is investigated further through dedicated diagnostic evaluations on the test set in Section 5.4.

5.4 Quantitative Results

This section reports final test-set results on the fixed AGCO test split (7 bags). I follow the standard depth-estimation protocol and report Abs Rel ($\downarrow$), Sq Rel ($\downarrow$), RMSE ($\downarrow$), RMSE log ($\downarrow$), and threshold accuracies δ_1 ($\uparrow$), δ_2 ($\uparrow$), and δ_3 ($\uparrow$). All numbers are taken from the consolidated experiment results sheet.

5.4.1 Baseline vs. Sparse LiDAR-Supervised Fine-tuning

Table 5.2 compares zero-shot evaluation against sparse LiDAR-supervised fine-tuning for three dataset fractions (20%, 50%, and 80%).

Zero-shot baseline. Without AGCO fine-tuning, DA-v3 provides the strongest baseline (Abs Rel = 0.323, RMSE = 1.370 m), followed by ZoeDepth (Abs Rel = 0.441, RMSE = 2.798 m), while MonoDepth2 performs worst (Abs Rel = 0.568, RMSE = 3.180 m). This is consistent with the fact that DA-v3 is trained with large-scale supervision/pseudo-supervision, whereas MonoDepth2’s original formulation is not designed to produce metric depth without adaptation.

Impact of sparse LiDAR fine-tuning. After sparse LiDAR-supervised fine-tuning, all models improve dramatically. For example, at 80% fine-tuning: DA-v3 achieves Abs Rel = 0.046 (RMSE = 0.503 m), MonoDepth2 reaches Abs Rel = 0.060 (RMSE = 0.618 m), and ZoeDepth reaches Abs Rel = 0.104 (RMSE = 1.879 m). This confirms that LiDAR supervision is highly effective for domain adaptation to agricultural scenes.

Effect of dataset fraction. Increasing the training fraction yields consistent gains for all models. DA-v3 improves from Abs Rel 0.055 $\rightarrow$ 0.049 $\rightarrow$ 0.046 as data increases from 20% to 80%. MonoDepth2 improves from 0.066 $\rightarrow$ 0.063 $\rightarrow$ 0.060, while ZoeDepth improves more moderately from 0.112 $\rightarrow$ 0.108 $\rightarrow$ 0.104. The accuracy thresholds (δ_1 , δ_2 , δ_3) also improve slightly with more data, indicating better overall consistency and fewer large depth failures.

Method	Abs Rel $\downarrow$	Sq Rel $\downarrow$	RMSE (m) $\downarrow$	RMSE log $\downarrow$	$\delta_1 \uparrow$	$\delta_2 \uparrow$	$\delta_3 \uparrow$
Monodepth2 (Zero-shot)	0.568	1.783	3.180	0.570	0.462	0.667	0.773
ZoeDepth (Zero-shot)	0.441	1.028	2.798	0.478	0.470	0.685	0.798
DA-v3 (Zero-shot, Large)	0.323	0.588	1.370	0.380	0.723	0.795	0.863
Monodepth2 finetune (20%)	0.066	0.067	0.664	0.118	0.953	0.983	0.994
ZoeDepth finetune (20%)	0.112	0.294	1.916	0.201	0.853	0.935	0.974
DA-v3 finetune (20%)	0.055	0.063	0.560	0.106	0.963	0.986	0.993
Monodepth2 finetune (50%)	0.063	0.072	0.635	0.123	0.961	0.981	0.991
ZoeDepth finetune (50%)	0.108	0.286	1.892	0.198	0.860	0.936	0.975
DA-v3 finetune (50%)	0.049	0.055	0.537	0.101	0.965	0.987	0.993
Monodepth2 finetune (80%)	0.060	0.071	0.618	0.121	0.963	0.980	0.990
ZoeDepth finetune (80%)	0.104	0.277	1.879	0.193	0.865	0.938	0.978
DA-v3 finetune (80%)	0.046	0.051	0.503	0.096	0.968	0.988	0.994

Table 5.2: Baseline (zero-shot) and sparse LiDAR-supervised finetuning results on the AGCO test set for three dataset fractions, where DA-v3 is the best performing model.

5.4.2 Fine-tuning on Smooth Depth and Evaluation on Smooth Depth

Table 5.3 reports performance when models are fine-tuned on the smooth supervision target and evaluated under the same smooth-depth evaluation protocol.

Monodepth2 remains stable. Monodepth2 achieves similar performance to sparse fine-tuning (e.g., Abs Rel $\approx$ 0.064–0.066 across fractions), suggesting that its training/evaluation pipeline is robust to the smooth ground-truth variant.

ZoeDepth and DA-v3 show inflated relative-error metrics. In contrast, ZoeDepth and DA-v3 report very large Abs Rel and Sq Rel values in this configuration (ZoeDepth Abs Rel ≈ 5.88 – 5.92 ; DA-v3 Abs Rel ≈ 4.50 – 4.90). Interestingly, δ accuracies remain relatively high (e.g., DA-v3 $\delta_1 \approx 0.93$ – 0.94), which indicates that many pixels remain within a multiplicative tolerance even while relative-error metrics explode. This combination strongly suggests that the smooth evaluation setting is sensitive to how filled pixels are treated (e.g., mask definition, depth scaling, or error normalization on filled regions).

Method (finetuned on smooth)	Abs Rel ↓	Sq Rel ↓	RMSE (m) ↓	RMSE log ↓	δ_1 ↑	δ_2 ↑	δ_3 ↑
Monodepth2 (20%)	0.066	0.067	0.621	0.120	0.953	0.982	0.993
ZoeDepth (20%)	5.869	51.397	2.398	0.441	0.811	0.907	0.956
DA-v3 (20%)	4.898	57.191	1.643	0.385	0.930	0.961	0.975
Monodepth2 (50%)	0.064	0.076	0.616	0.123	0.960	0.980	0.990
ZoeDepth (50%)	5.922	52.015	2.374	0.440	0.815	0.909	0.957
DA-v3 (50%)	4.772	57.940	1.624	0.383	0.935	0.954	0.976
Monodepth2 (80%)	0.065	0.074	0.598	0.124	0.960	0.980	0.989
ZoeDepth (80%)	5.882	51.952	2.375	0.438	0.818	0.908	0.958
DA-v3 (80%)	4.498	55.449	1.627	0.375	0.933	0.964	0.975

Table 5.3: Results when finetuning and evaluating using the smooth (dense) depth configuration, where the best-performing model is Monodepth2

5.4.3 Diagnostic Evaluation: Smooth-trained Models Evaluated on LiDAR-GT Pixels

To separate training effects from evaluation effects, Table 5.4 reports results where models are trained on smooth targets but evaluated only on LiDAR-valid pixels.

Method (finetuned on smooth, eval on LiDAR-GT)	Abs Rel ↓	Sq Rel ↓	RMSE (m) ↓	RMSE log ↓	δ_1 ↑	δ_2 ↑	δ_3 ↑
Monodepth2 (20%)	0.066	0.069	0.665	0.120	0.953	0.982	0.993
ZoeDepth (20%)	0.113	0.295	1.919	0.201	0.852	0.935	0.975
DA-v3 (20%)	0.060	0.074	0.687	0.118	0.963	0.983	0.992
Monodepth2 (50%)	0.064	0.078	0.667	0.124	0.959	0.980	0.990
ZoeDepth (50%)	0.114	0.286	1.891	0.199	0.856	0.936	0.976
DA-v3 (50%)	0.052	0.060	0.635	0.106	0.966	0.986	0.993
Monodepth2 (80%)	0.065	0.076	0.650	0.124	0.960	0.979	0.989
ZoeDepth (80%)	0.107	0.283	1.892	0.197	0.859	0.936	0.977
DA-v3 (80%)	0.056	0.064	0.643	0.110	0.963	0.986	0.992

Table 5.4: Diagnostic evaluation: models finetuned on smooth targets, evaluated only on LiDAR-GT valid pixels, where DA-v3 is showing the best performance.

Under this LiDAR-GT evaluation, ZoeDepth and DA-v3 return to expected performance ranges (e.g., ZoeDepth Abs Rel ≈ 0.107 – 0.114 and DA-v3 Abs Rel ≈ 0.052 – 0.060 de-

pending on fraction). This indicates that smooth fine-tuning does not necessarily destroy performance on real LiDAR measurements; instead, the extreme errors mainly arise from the smooth evaluation protocol itself.

5.4.4 Diagnostic Evaluation: Cropped-Region (“1.4/3 of the image”)

Finally, Table 5.5 reports the cropped evaluation (“1.4/3 of the image”) listed in the results sheet. In this setting, Abs Rel and Sq Rel become unusually large for all models (e.g., Monodepth2 Abs Rel ≈ 0.99 –1.03), while δ accuracies remain high for DA-v3 fine-tuned models. This again suggests a strong interaction between the evaluation region/mask and the behaviour of relative-error metrics.

Method (smooth finetune, eval on “1.4/3” crop)	Abs Rel ↓	Sq Rel ↓	RMSE (m) ↓	RMSE log ↓	δ_1 ↑	δ_2 ↑	δ_3 ↑
Monodepth2 (20%)	0.989	12.288	0.696	0.167	0.953	0.981	0.991
ZoeDepth (20%)	1.621	9.886	1.103	0.309	0.897	0.952	0.975
DA-v3 (20%)	1.120	9.812	0.617	0.226	0.966	0.980	0.990
Monodepth2 (50%)	0.994	12.426	0.710	0.170	0.957	0.978	0.989
ZoeDepth (50%)	1.674	10.383	1.058	0.309	0.900	0.954	0.976
DA-v3 (50%)	1.112	10.803	0.578	0.218	0.968	0.984	0.991
Monodepth2 (80%)	1.026	13.353	0.698	0.171	0.958	0.977	0.988
ZoeDepth (80%)	1.617	10.240	1.064	0.306	0.903	0.953	0.977
DA-v3 (80%)	0.978	10.370	0.597	0.197	0.963	0.984	0.990

Table 5.5: Diagnostic evaluation of models fine-tuned on smooth targets and evaluated on a cropped region (“1.4/3 of the image”), where on the basis of all three accuracy DA-v3 is doing the best performance.

5.5 Model Comparison and Discussion

This section compares Monodepth2, ZoeDepth, and Depth Anything v3 (DA-v3) by jointly analyzing quantitative results (Section 5.4) and qualitative depth predictions. The goal is to understand how different training paradigms and architectural choices affect performance under LiDAR-supervised fine-tuning, and to explain the trends observed across multiple evaluation settings.

For each model, qualitative examples are presented that show zero-shot predictions alongside predictions obtained after sparse and smooth LiDAR-supervised fine-tuning. These visual comparisons help explain the numerical trends reported earlier and provide insight into model behaviour in agricultural environments.

5.5.1 Qualitative Comparison: Zero-shot vs Fine-tuned Predictions

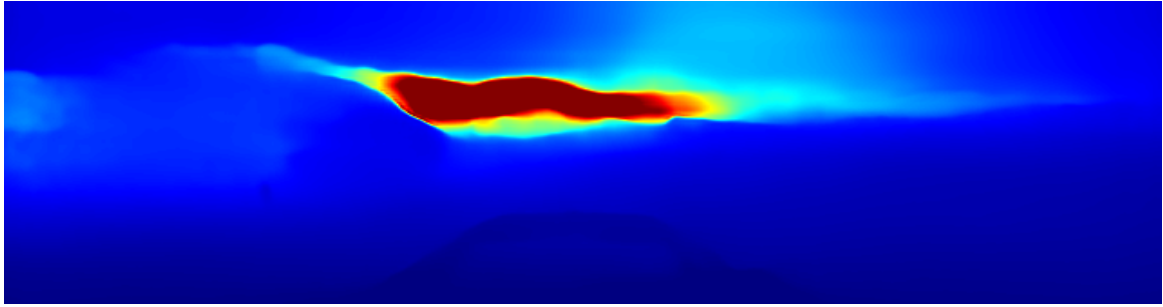
Across all three models, a clear qualitative difference is observed between zero-shot predictions and fine-tuned predictions. In the zero-shot (baseline) setting, depth maps often

exhibit scale inconsistency, incorrect depth ordering, and reduced sensitivity to scene geometry specific to agricultural environments. Figure 5.7 illustrates typical zero-shot behaviour across the three models, highlighting scale inconsistency and weaker adaptation to agricultural geometry. After fine-tuning with LiDAR supervision, predictions become more geometrically consistent, especially in the near-field region that is most relevant for safety.

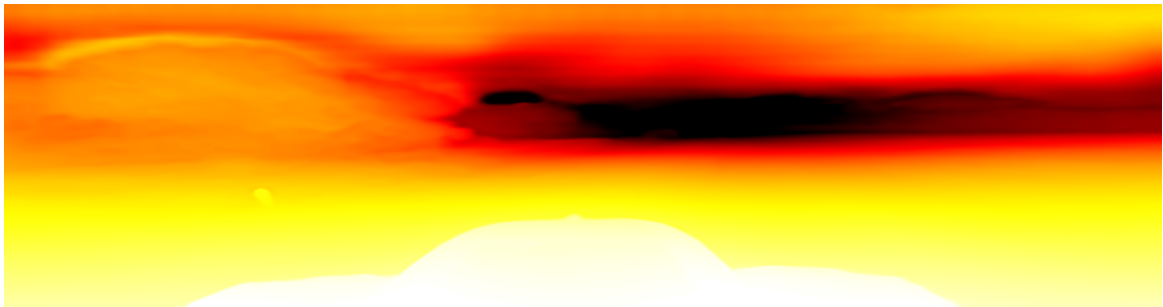
Sparse fine-tuning generally produces sharp and physically meaningful depth structures at locations where LiDAR supervision is available. Figure 5.8 shows that sparse LiDAR fine-tuning produces more coherent near-field geometry and clearer obstacle boundaries across all models. Smooth fine-tuning produces visually denser depth maps, but its effect on metric accuracy varies significantly across models. Figure 5.9 demonstrates that smooth supervision yields visually denser predictions, while the quantitative behaviour depends strongly on how filled pixels are treated during evaluation.



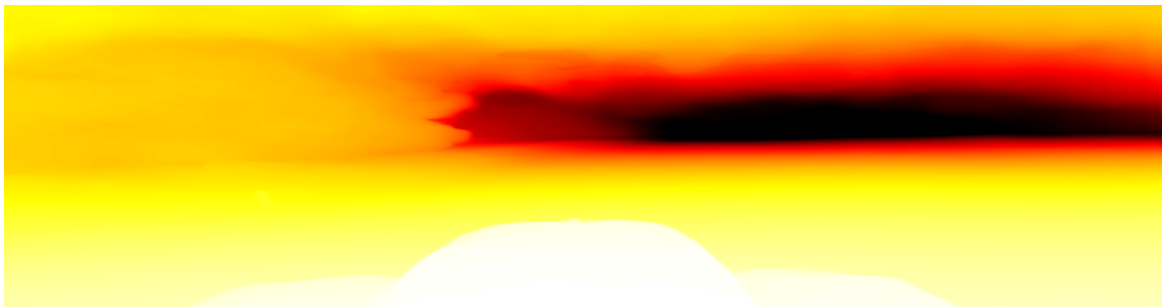
RGB input



Monodepth2 (zero-shot)



ZoeDepth (zero-shot)



Depth Anything v3 (zero-shot)

Figure 5.7: Qualitative comparison in the zero-shot (baseline) setting. Depth predictions are generated using pretrained checkpoints without fine-tuning.

5.5.2 Model-Specific Behaviour

5.5.2.1 Monodepth2

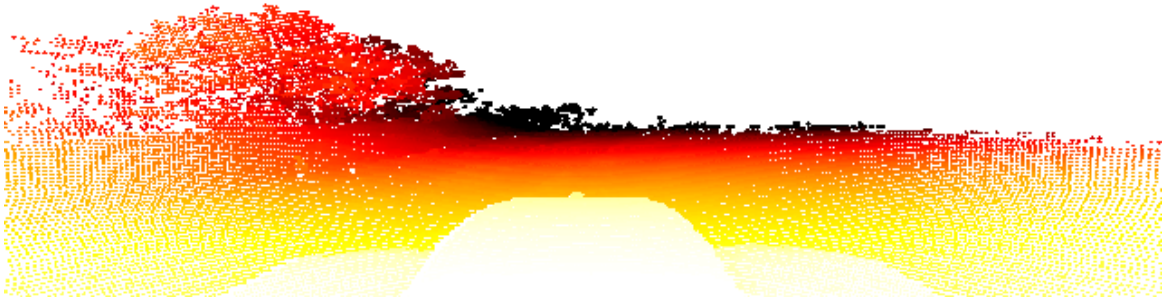
Monodepth2 shows the largest qualitative improvement from zero-shot to fine-tuned settings. In the zero-shot case, depth predictions suffer from scale ambiguity and often compress the depth range, which is consistent with its self-supervised training paradigm.

After sparse LiDAR fine-tuning, Monodepth2 produces stable and coherent depth maps with accurate near-field geometry. Ground surfaces become smoother and obstacle boundaries are better defined. Importantly, Monodepth2 remains stable under smooth supervision, both visually and numerically. This stability is reflected in the quantitative results, where Monodepth2 maintains similar Abs Rel and RMSE values under sparse and smooth evaluation.

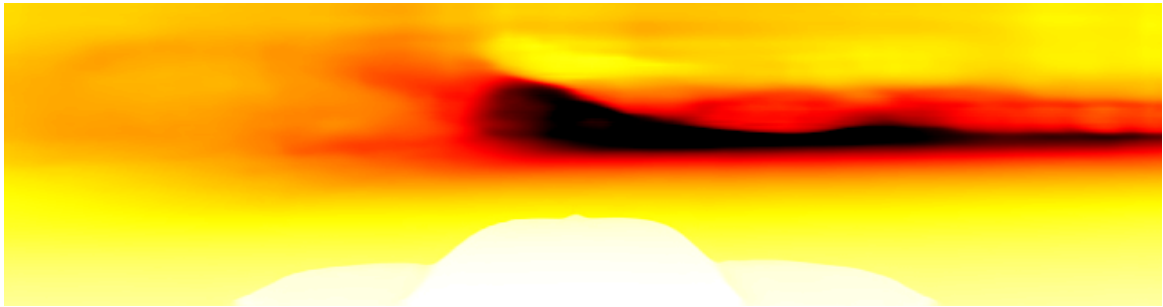
This behaviour suggests that Monodepth2 benefits strongly from explicit metric supervision and adapts well to different ground-truth density configurations. However, its overall accuracy remains slightly lower than DA-v3 in the best-performing sparse setting.



RGB input



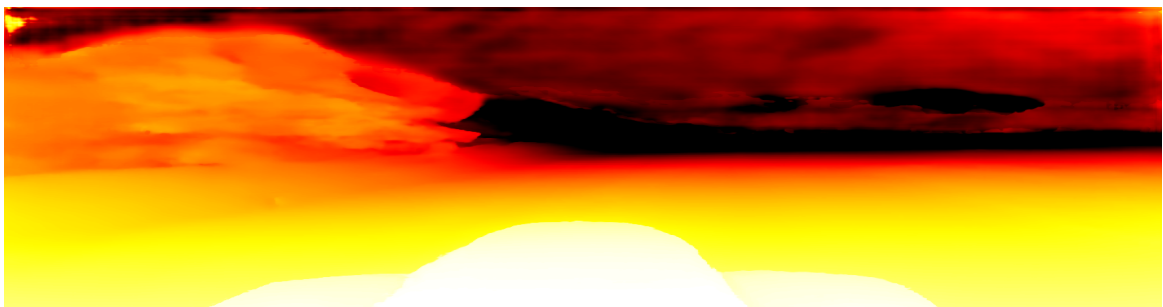
Sparse LiDAR ground truth



Monodepth2 (sparse fine-tuned)



ZoeDepth (sparse fine-tuned)



Depth Anything v3 (sparse fine-tuned)

Figure 5.8: Qualitative comparison after sparse LiDAR-supervised fine-tuning. From top to bottom: rectified RGB input, sparse LiDAR ground-truth depth, and depth predictions from Monodepth2, ZoeDepth, and Depth Anything v3 for the same test sample.

5.5.2.2 ZoeDepth

ZoeDepth starts from a stronger zero-shot baseline than Monodepth2, producing more reasonable metric depth without fine-tuning. However, its fine-tuned performance improves more gradually and remains consistently behind DA-v3 and Monodepth2 in sparse LiDAR-supervised experiments.

Qualitatively, sparse fine-tuning improves depth ordering and near-field consistency, but predictions remain smoother and less detailed than those of DA-v3. Under smooth supervision, ZoeDepth produces visually plausible depth maps, but quantitative metrics degrade significantly when evaluated using the smooth ground-truth configuration.

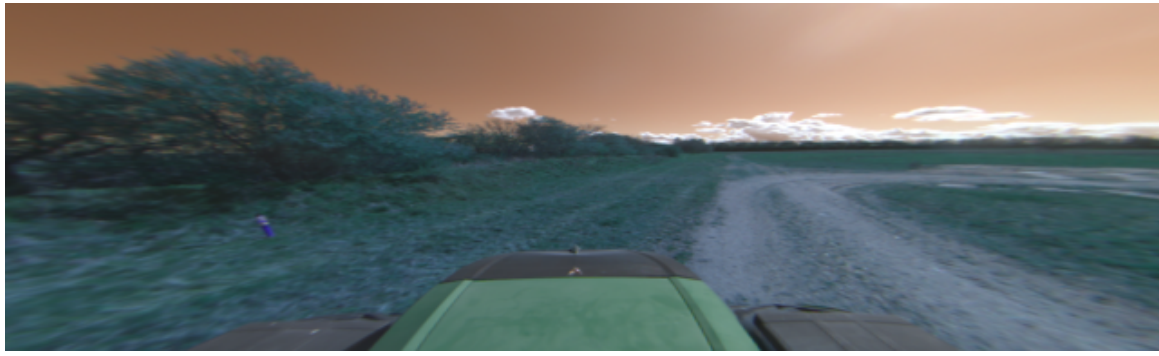
This discrepancy between visual quality and metric performance highlights a key limitation, ZoeDepth’s relative-error metrics are highly sensitive to filled depth values introduced during smoothing. When evaluation is restricted back to original LiDAR-valid pixels, performance returns to expected levels, confirming that the issue lies primarily in the evaluation interaction rather than in complete model failure.

5.5.2.3 Depth Anything v3

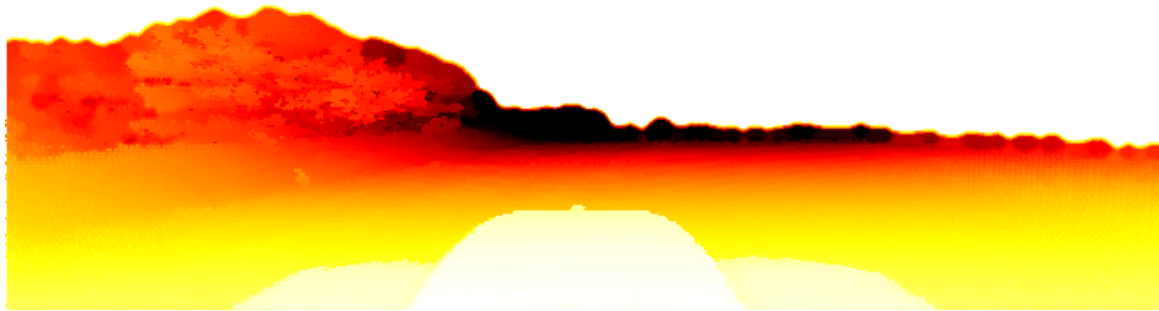
Depth Anything v3 consistently delivers the strongest overall performance after sparse LiDAR fine-tuning. Qualitatively, it produces the most detailed and spatially consistent depth maps, with clear object boundaries and stable ground-plane estimation.

Zero-shot predictions from DA-v3 are already strong, and fine-tuning further improves near-field accuracy and reduces scale bias. This is reflected in the lowest Abs Rel and RMSE values across all sparse fine-tuning fractions.

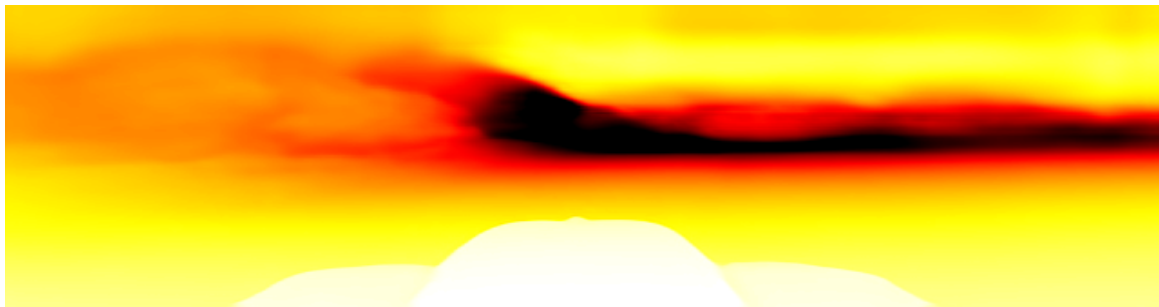
However, similar to ZoeDepth, DA-v3 exhibits inflated relative-error metrics under smooth evaluation. Despite this, qualitative inspection shows that predicted depth maps remain reasonable, and diagnostic evaluations confirm strong performance when evaluation is restricted to LiDAR-valid pixels. This again indicates a mismatch between smooth ground-truth evaluation and relative-error metrics rather than a breakdown in learned geometry.



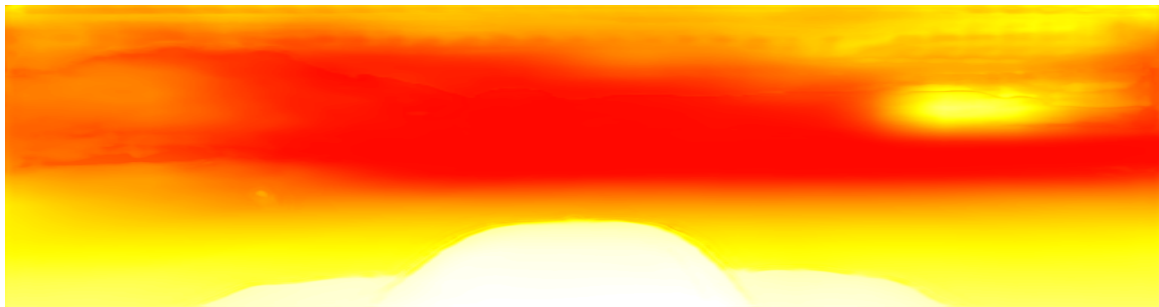
RGB input



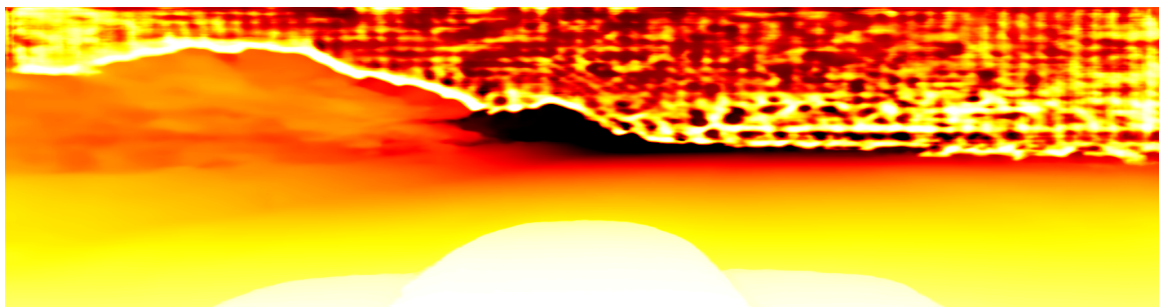
Smoothed LiDAR ground truth



Monodepth2 (smooth fine-tuned)



ZoeDepth (smooth fine-tuned)



Depth Anything v3 (smooth fine-tuned)

Figure 5.9: Qualitative comparison after smooth LiDAR-supervised fine-tuning. From top to bottom: rectified RGB input, smoothed LiDAR-based ground-truth depth, and depth predictions from Monodepth2, ZoeDepth, and Depth Anything v3 for the same test sample.

5.5.3 Cross-Model Trends

Several consistent trends emerge across models:

- **Sparse LiDAR supervision is highly effective:** All models benefit significantly from sparse fine-tuning, demonstrating that even limited LiDAR supervision is sufficient for strong domain adaptation.
- **DA-v3 achieves the best overall performance:** Both quantitatively and qualitatively, DA-v3 provides the most accurate and consistent depth predictions under sparse supervision.
- **Monodepth2 shows exceptional robustness:** While not the most accurate, Monodepth2 remains stable across all evaluation configurations, including smooth supervision.
- **Smooth supervision introduces evaluation sensitivity:** For ZoeDepth and DA-v3, smooth ground-truth evaluation leads to extreme relative-error values despite visually plausible predictions.

These trends highlight the importance of aligning ground-truth construction and evaluation protocols with the intended use of the model.

5.5.4 Error Analysis and Experimental Design Choices

5.5.4.1 Why Use a Cropped Evaluation Region (1.4/3 of the Image)

The cropped evaluation setting (1.4/3 of the image) is introduced to reduce the influence of regions that are less relevant for safety-oriented depth reasoning. In agricultural scenes, the upper part of the image often contains sky, tree canopies, or distant background elements with sparse or unreliable LiDAR coverage.

By focusing evaluation on the lower portion of the image, the analysis emphasizes ground surfaces and near-field obstacles, which are more critical for proximity estimation and braking decisions. The resulting metric changes observed in the cropped evaluation further illustrate how depth error distributions are highly dependent on the spatial region being evaluated.

5.5.4.2 Why Perform Multiple Evaluation Configurations

Multiple evaluation configurations are used to disentangle different sources of error:

- **Sparse evaluation** reflects true LiDAR measurement reliability.

- **Smooth evaluation** tests robustness under denser supervision but reveals sensitivity in relative-error metrics.
- **LiDAR-GT evaluation after smooth training** isolates training effects from evaluation artifacts.
- **Cropped evaluation** focuses on safety-relevant regions.

Together, these experiments provide a more complete picture of model behaviour than a single metric configuration would allow.

5.5.4.3 Implications for Practical Use

From a practical perspective, the results suggest that sparse LiDAR-supervised fine-tuning combined with careful evaluation masking provides the most reliable assessment of monocular depth models for agricultural safety. DA-v3 emerges as the strongest candidate for deployment, while Monodepth2 offers robustness and simplicity. Smooth supervision can be useful for visual consistency but must be handled carefully to avoid misleading quantitative conclusions.

Overall, this comparison demonstrates that model performance cannot be judged by metrics alone; qualitative inspection, evaluation design, and domain-specific reasoning are essential for meaningful depth estimation analysis.

6 Conclusion and Future Work

This thesis investigated LiDAR-supervised monocular depth estimation for agricultural machinery, with a focus on building a reproducible dataset pipeline from raw ROS bag recordings and benchmarking modern depth models under controlled conditions.

Answers to Research Questions

- **How does the dataset need to be prepared?**

Raw LiDAR-camera recordings require rectification, synchronization, LiDAR-camera projection, depth map generation, and validity masking to produce reliable RGB-depth pairs suitable for supervised learning.

- **How can LiDAR points be projected and visualized over RGB images?**

LiDAR points are transformed into the camera frame using extrinsics and projected using the pinhole camera model, with Z-buffering to preserve the nearest visible surface.

- **What should be used as ground-truth depth for training and evaluation?**

This work used LiDAR-projected sparse depth as the primary supervision signal and introduced a controlled smoothed variant to study the effect of denser supervision while preserving measured values.

- **Can an automated pipeline reduce manual work while still producing reliable results?**

The developed pipeline is largely automated, but practical recordings can still require limited manual extrinsic refinement to correct small misalignments and ensure accurate projection.

- **Which models perform best under LiDAR-supervised fine-tuning in this setting?**

Across experiments, increasing training data generally improved stability and performance. Depth Anything v3 showed the strongest and most consistent convergence under sparse supervision in this agricultural domain.

6.1 Main Contributions

- A reproducible dataset preparation pipeline converting raw ROS recordings into rectified RGB images paired with LiDAR-derived depth maps.

- Bag-level splitting with a fixed test set to avoid temporal leakage and support fair benchmarking.
- Two ground-truth variants (sparse and smoothed) enabling controlled analysis of supervision density and evaluation effects.
- A unified benchmarking setup for MonoDepth2, ZoeDepth, and Depth Anything v3 under identical splits and metrics.

6.2 Key Findings and Limitations

LiDAR-supervised fine-tuning significantly improves monocular depth estimation performance compared to zero-shot evaluation across all models. Larger training fractions lead to smoother training behaviour, more stable validation trends, and consistent improvements on the fixed AGCO test set.

Depth Anything v3 achieves the strongest overall performance under sparse LiDAR supervision, while Monodepth2 shows the most robust behaviour across different supervision and evaluation settings. ZoeDepth benefits from LiDAR supervision but is more sensitive to ground-truth density and evaluation design.

A key finding is that evaluation protocol strongly influences reported performance when smoothed ground truth is used. For ZoeDepth and Depth Anything v3, relative-error metrics become inflated when filled depth pixels are included in evaluation, despite visually reasonable predictions. Diagnostic experiments indicate that this effect is mainly caused by metric sensitivity rather than a failure of the models.

Limitations include residual dependence on manual extrinsic refinement, inherent sparsity of LiDAR ground truth, and sensitivity of some evaluation metrics to the choice of evaluation mask and depth representation.

6.3 Future Work

This thesis establishes a strong dataset and benchmarking foundation. The following directions would strengthen reliability to move toward safety-oriented deployment.

6.3.1 Downstream Use Case: Depth-Based Proximity Cues (Concept)

It is included as future work to illustrate how the trained monocular depth models could be extended toward safety-oriented perception in agricultural scenes.

Many monocular distance estimation approaches follow a simple idea: object detection tells *where* an object is in the image, and depth estimation tells *how far* it is from the camera. By combining these two pieces of information, a monocular system can estimate the absolute distance to objects using only a single RGB image [4, 5].

A simplified extension is to explore a *depth-only* cue that does not rely on explicit object detection. The idea is to define a forward-driving region of interest (ROI), for example a rectangle or trapezoid corridor directly in front of the vehicle, and monitor predicted depth structure inside this ROI. Proximity can be estimated using robust statistics over the closest valid depths (e.g., near-percentile or median over near-depth pixels) instead of a raw minimum, which reduces sensitivity to noise and isolated outliers.

Although a depth-only approach cannot reliably assign semantics (e.g., *person* vs. *tree*), it can still provide a conservative obstacle likelihood signal. For instance, near obstacles often appear as coherent connected regions that interrupt the smoother ground-depth trend, while vegetation may produce more fragmented patterns with higher local depth variance. With known camera mounting and scene geometry, a further plausibility check can be added: for a given estimated distance, a standing person is expected to occupy a reasonable vertical extent in the image. This does not guarantee classification, but it can bias the likelihood score toward potentially human-sized obstacles in the near field.

6.3.2 Downstream Use Case: TTC Estimation from Depth (Concept)

Once an obstacle distance d is estimated from the ROI, a basic time-to-collision (TTC) signal can be derived using the vehicle speed v (from odometry or onboard sensors) as $TTC \approx d/v$ under a constant-speed assumption.

As a simple example, at 20 km/h (≈ 5.56 m/s), an obstacle at 20 m corresponds to a TTC of about 3.6 s, while 25 m corresponds to about 4.5 s. Incorporating a conservative reaction/actuation delay (e.g., 1 s) further motivates early thresholds, because the vehicle would travel about 5.56 m during that delay, reducing the remaining safety margin before braking begins.

Future work should evaluate these depth-driven proximity and TTC cues on recorded sequences with near-field obstacle events and compare them against detector+depth baselines under challenging conditions such as vegetation, shadows, and partial occlusions.

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